CS-649: Cloud DevOps

LAB 02

Hands-On Git and Version Control

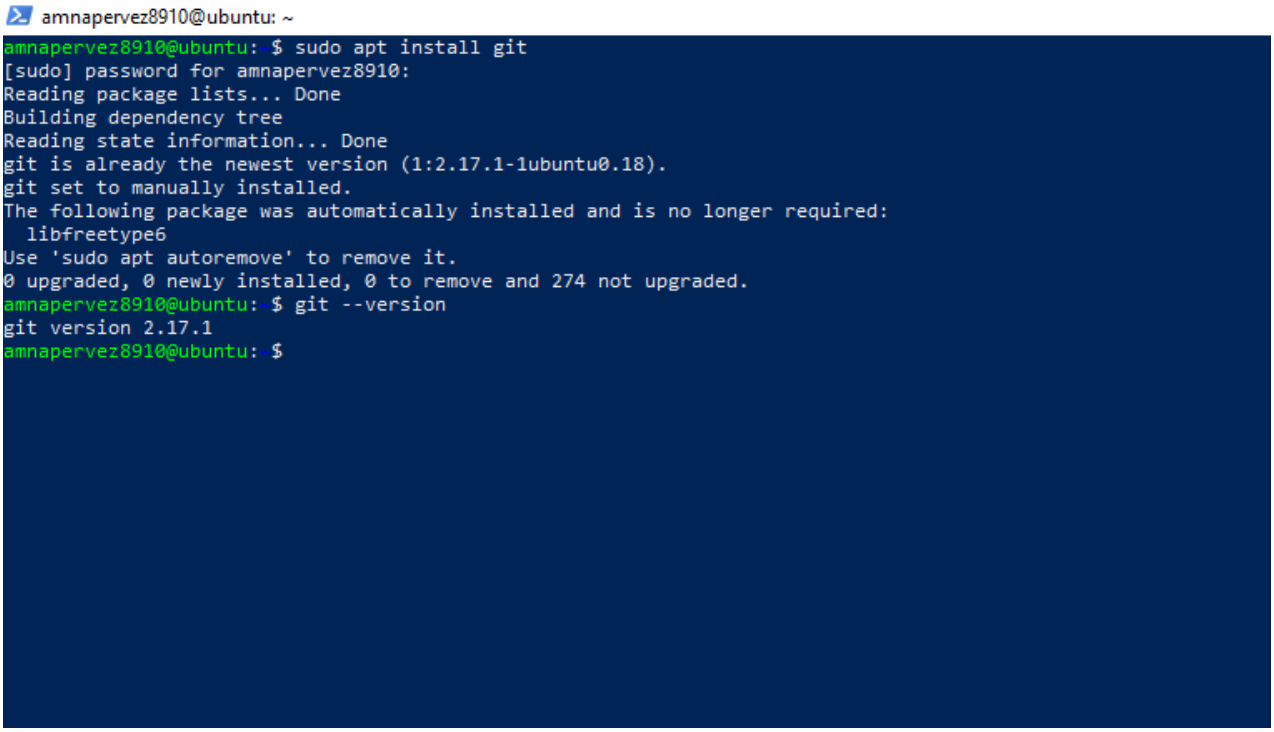
Student Name	**Amna Pervez**
Registration No.	**2023-BSCS-10**
Section	**A**
GitHub Username	**amnapervez8910**

Practical Work and Screenshot Evidence

Tasks 1–13 and Q1–Q3

Getting Started

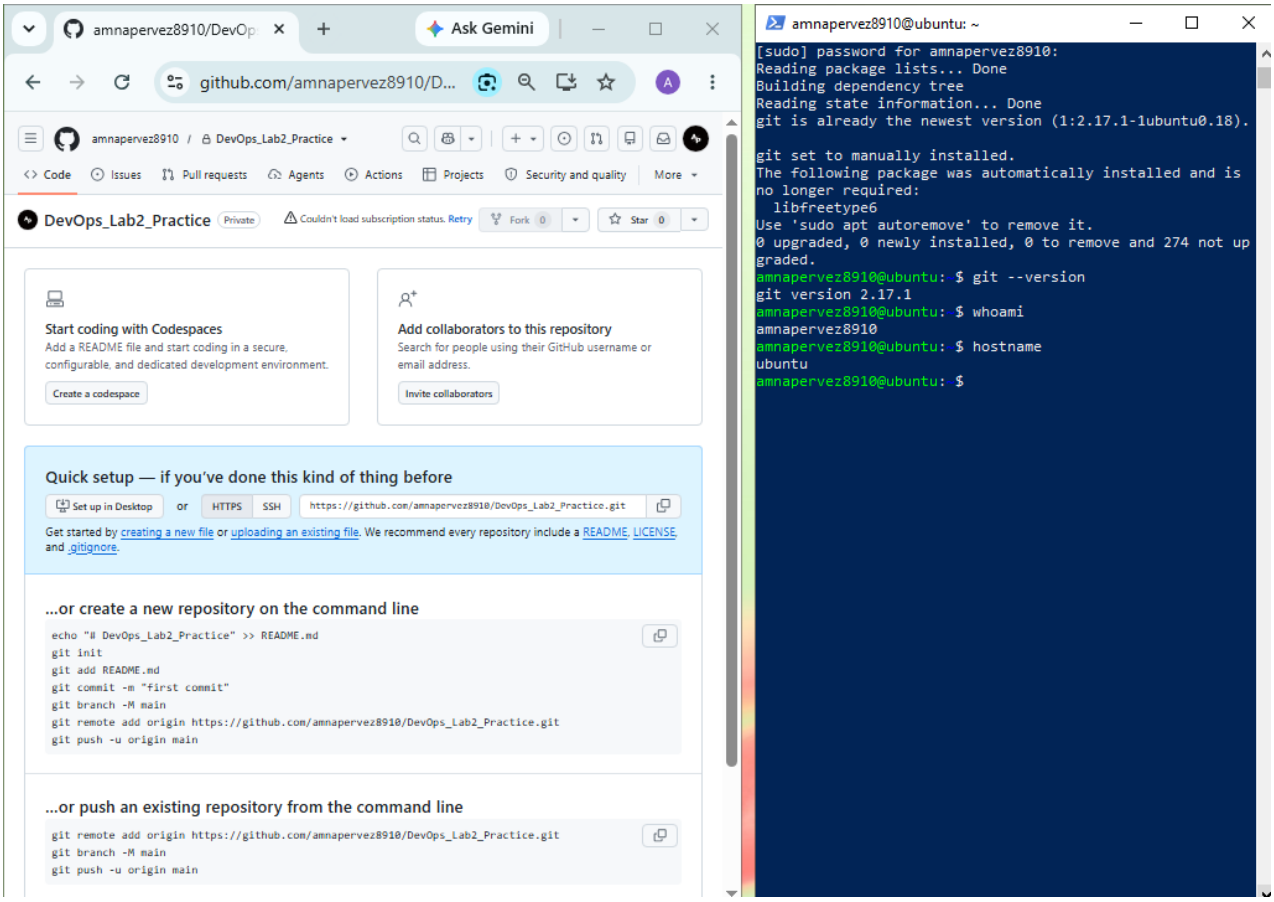
git_installation.png



Git installation checked on Ubuntu; installed Git version verified.

Task 1: Private GitHub Repository

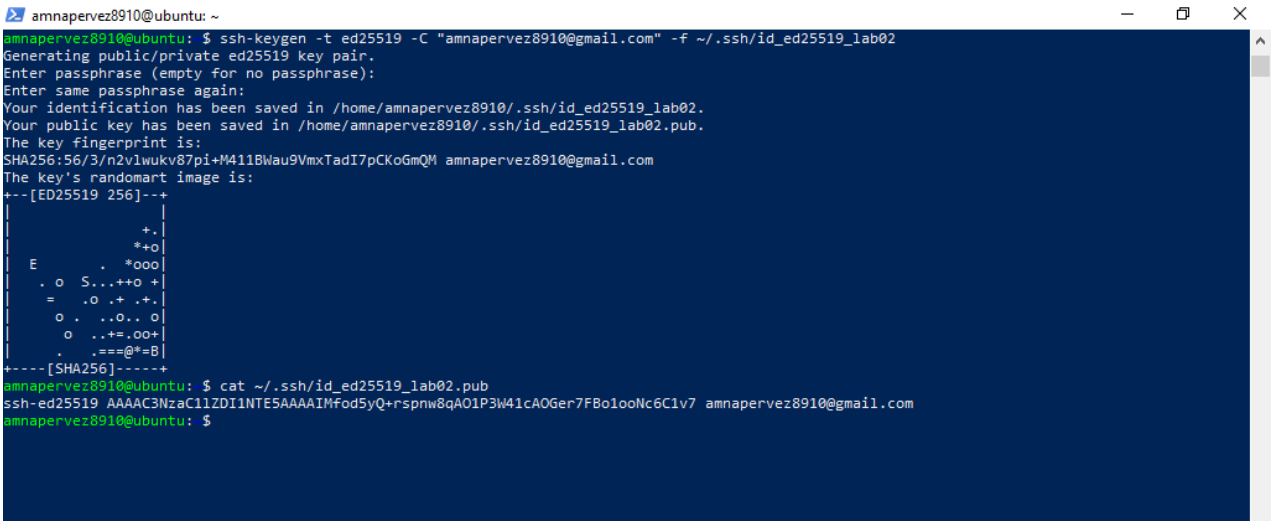
repo_private.png



Private DevOps_Lab2_Practice repository shown with the required Ubuntu identity.

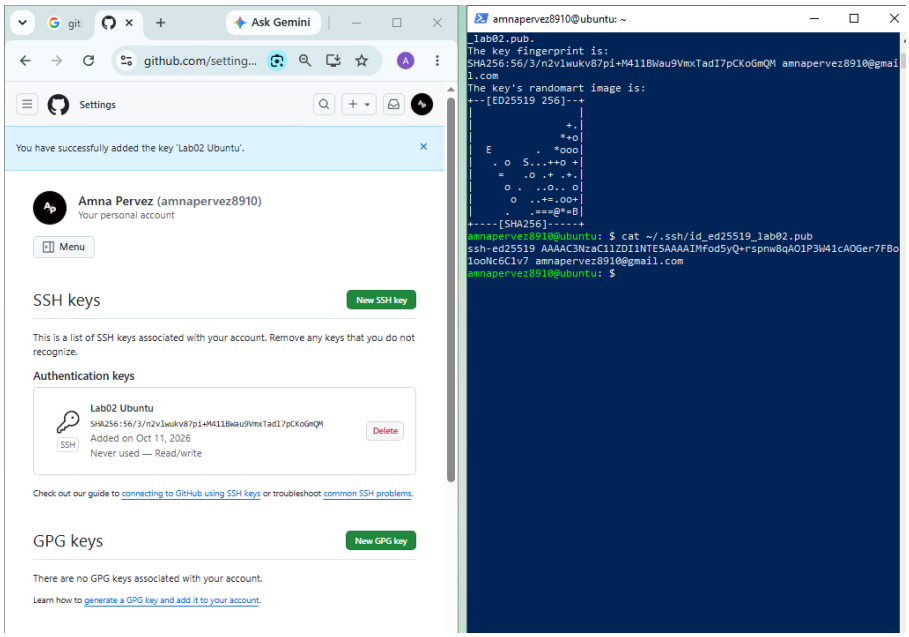
Task 2: Connect via SSH

ssh_keygen.png



Ed25519 key generated in Ubuntu; only the public key is displayed.

github_sshkey.png



Ubuntu public key added to GitHub as Lab02 Ubuntu.

Task 2: Connect via SSH (continued)

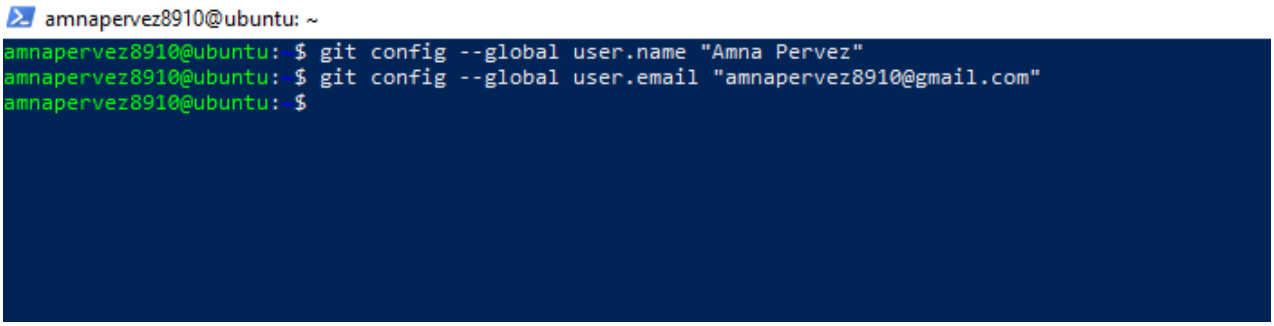
ssh_clone.png



Separate SSH verification clone of the completed practice repository; all objects received successfully.

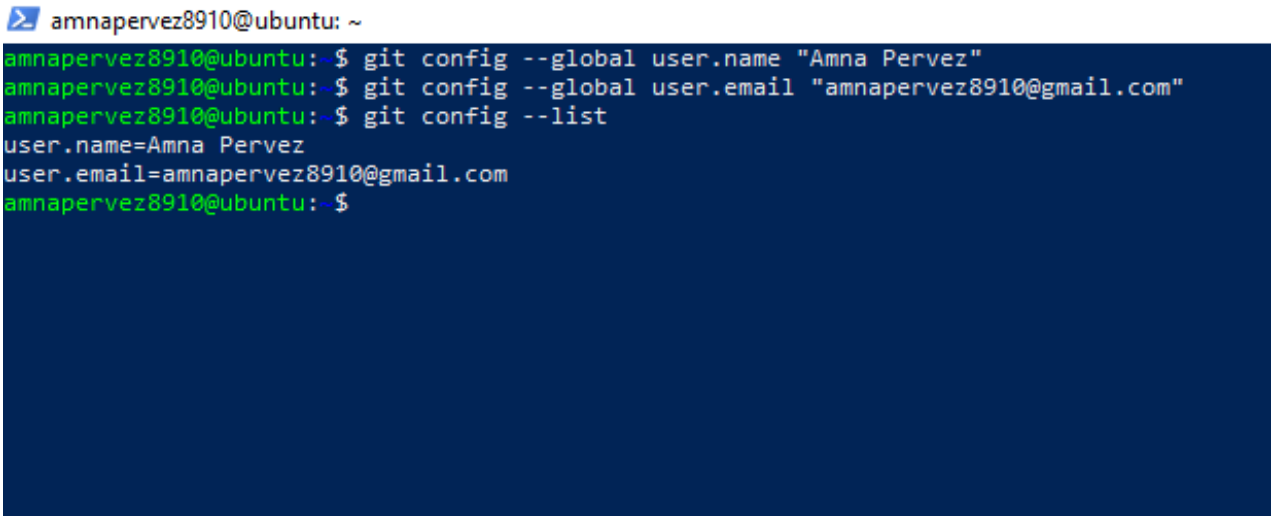
Task 3: Git Identity

git_identity.png



Git author name and email configured in Ubuntu.

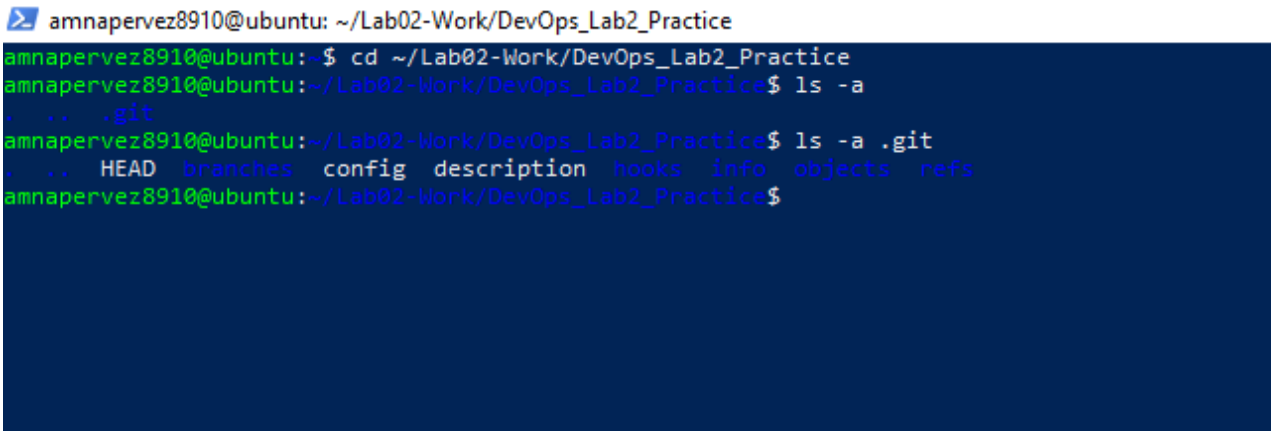
git_config_list.png



Git configuration listed to verify the author name and email.

Task 4: Explore the .git Folder

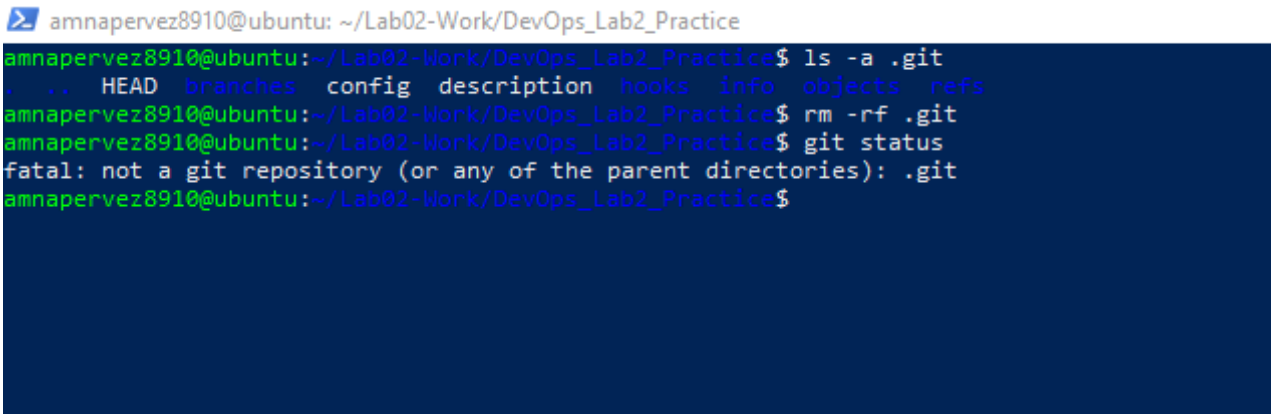
git_folder.png



Hidden .git directory and its internal files listed.

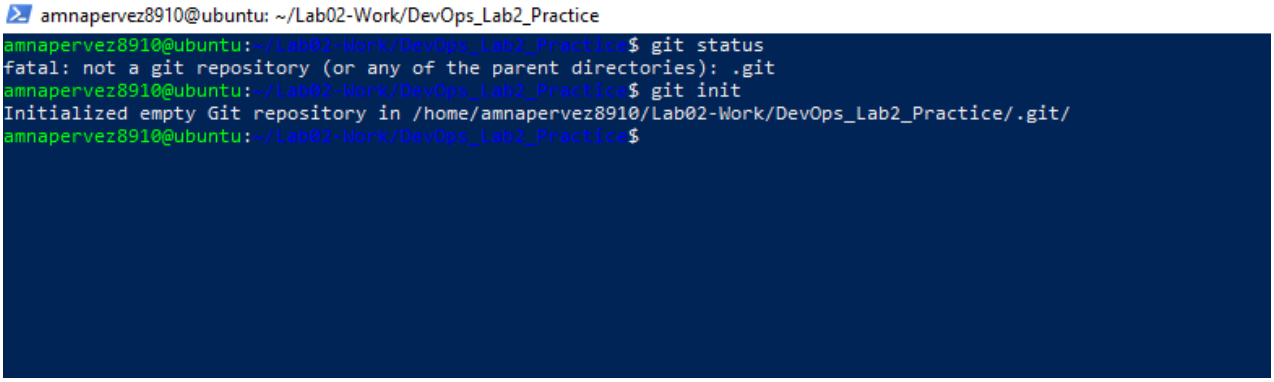
Task 5: Local Repository Management

delete_git.png



Practice-copy .git metadata removed; git status reports the expected non-repository error.

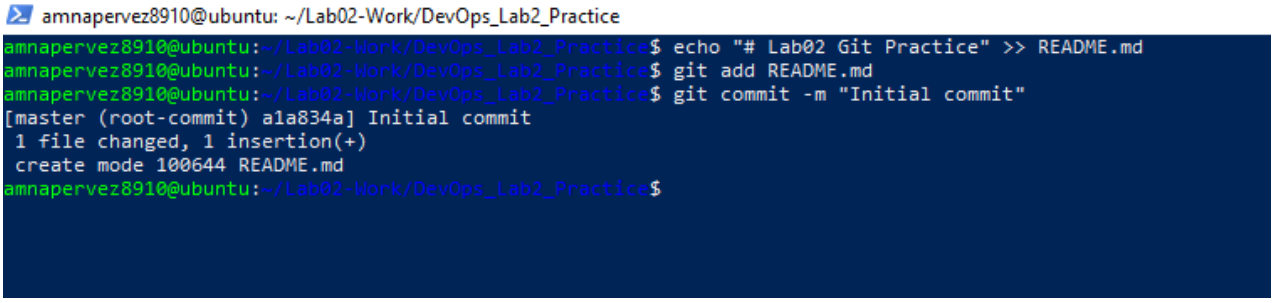
git_init.png



Local practice repository re-initialized with git init.

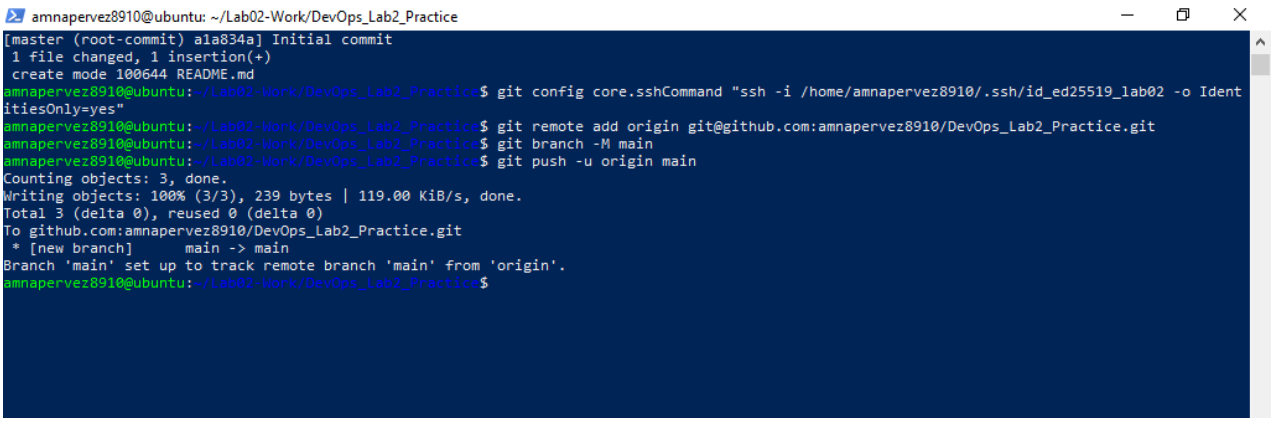
Task 5: Local Repository Management (continued)

first_commit.png



README.md staged and committed as the initial commit.

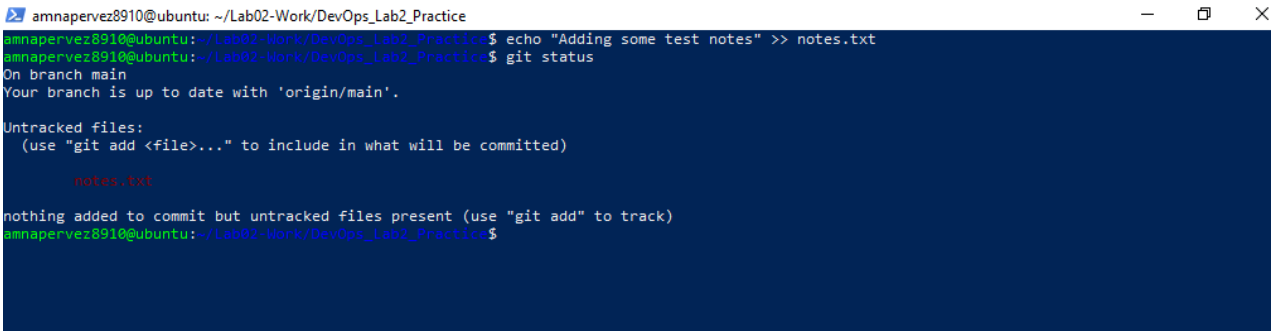
first_push.png



SSH origin configured, branch renamed main, and initial commit pushed.

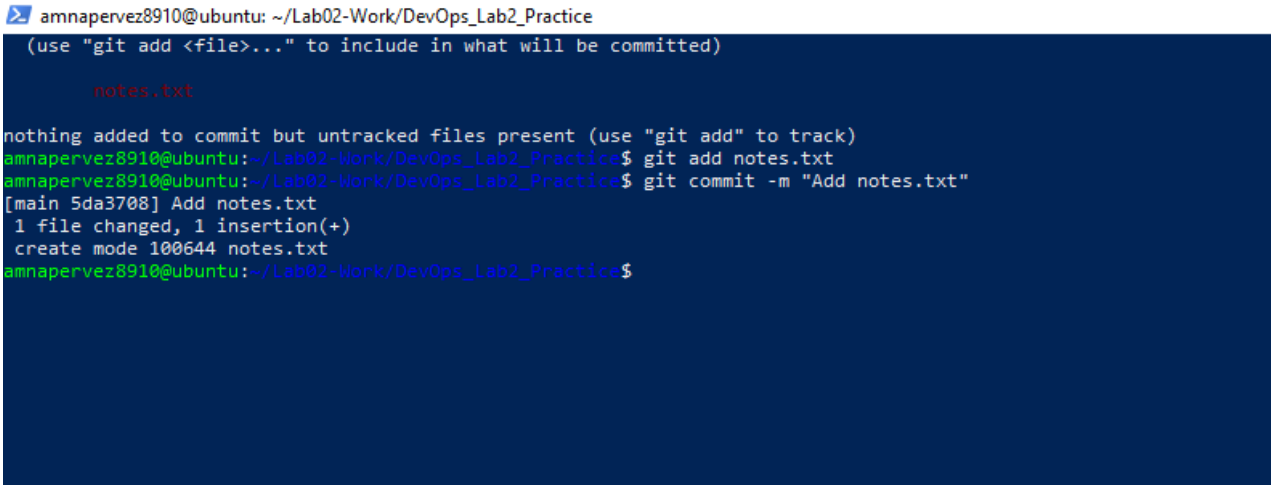
Task 6: File Status and Staging

status1.png



New notes.txt file shown as untracked.

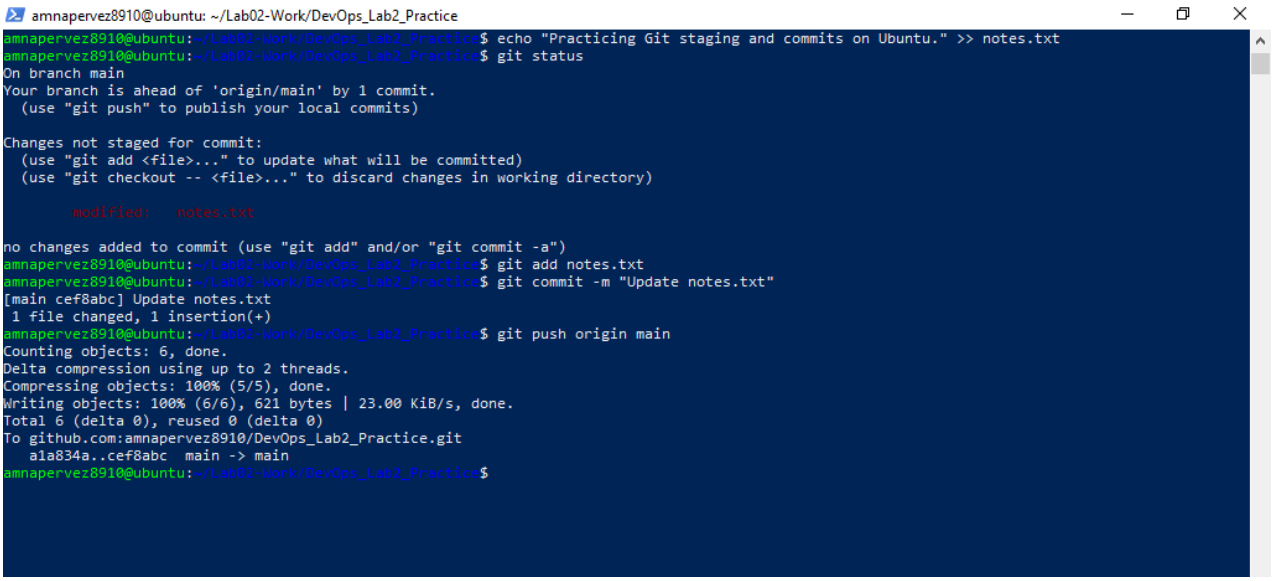
commit_notes.png



notes.txt staged and committed.

Task 6: File Status and Staging (continued)

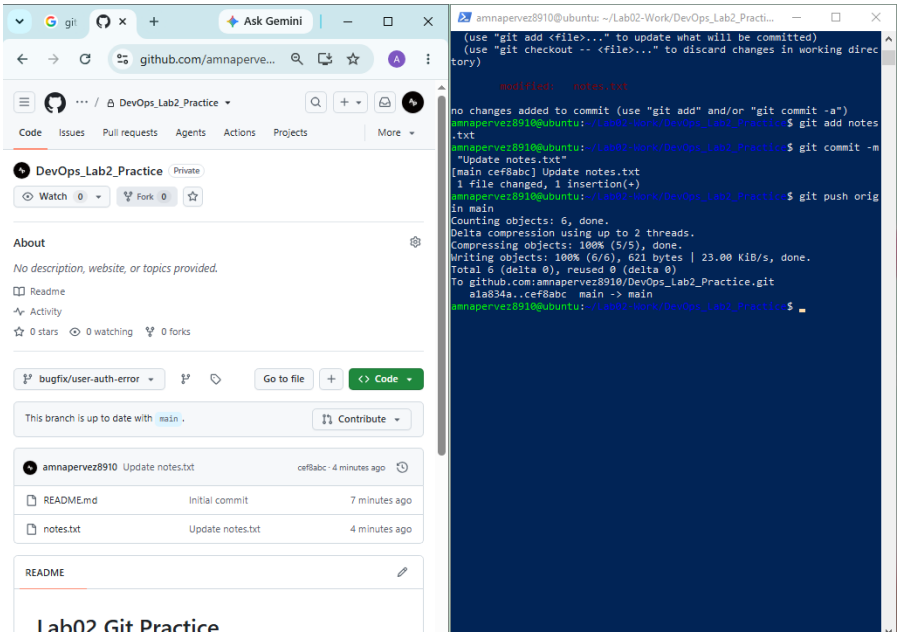
notes_update.png



notes.txt edited, committed again, and pushed to main.

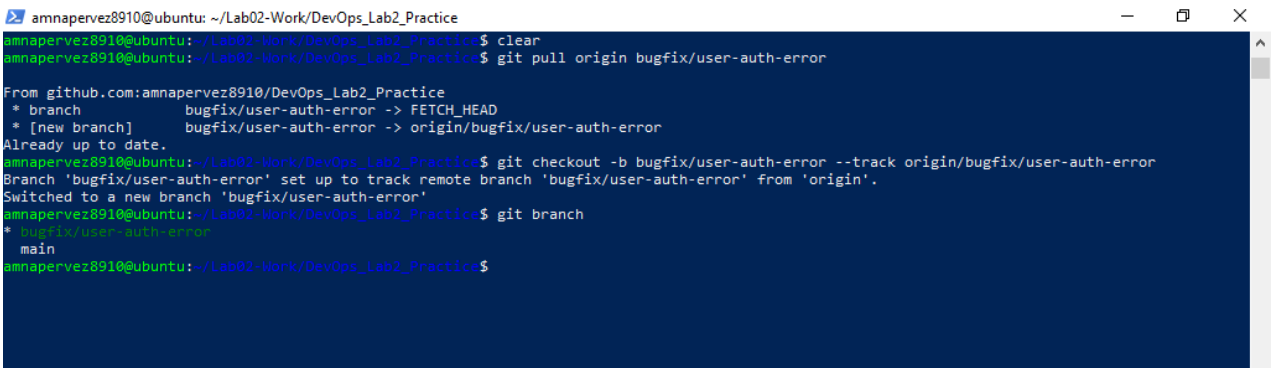
Task 7: Branch Creation in GitHub

bugfix_branch_gui.png



bugfix/user-auth-error branch created through the GitHub interface.

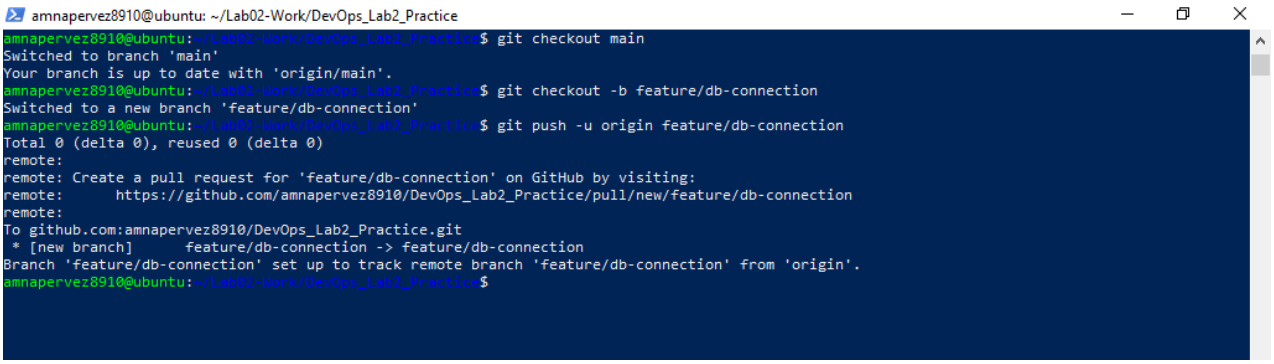
bugfix_branch_local.png



Remote bugfix branch pulled, local tracking branch created, and branch list displayed.

Task 8: Feature Branch Creation and Push

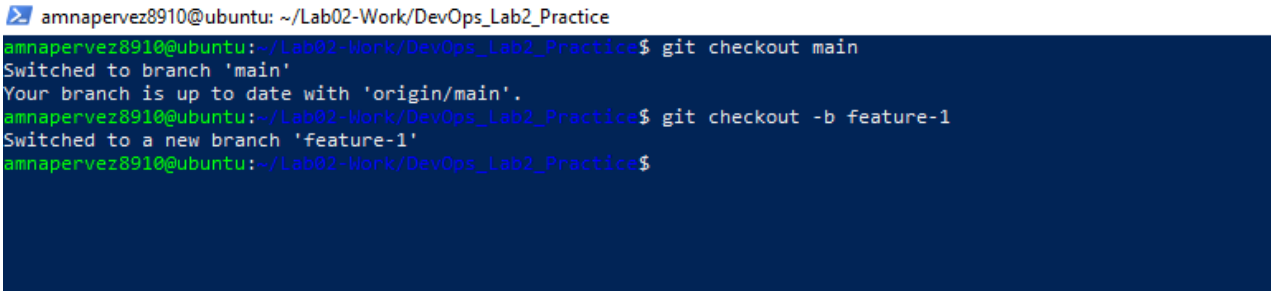
feature_db_branch.png



feature/db-connection branch created and pushed from the Ubuntu terminal.

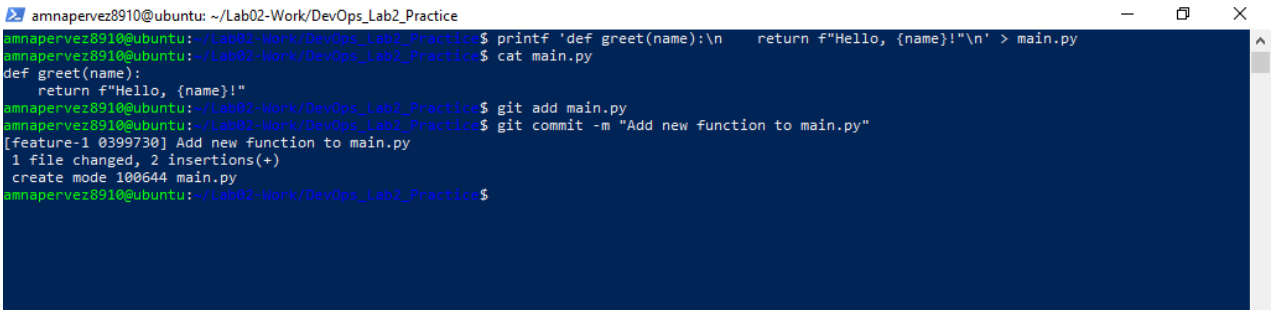
Task 9: Branching and Merging

branch_create.png



feature-1 branch created from main.

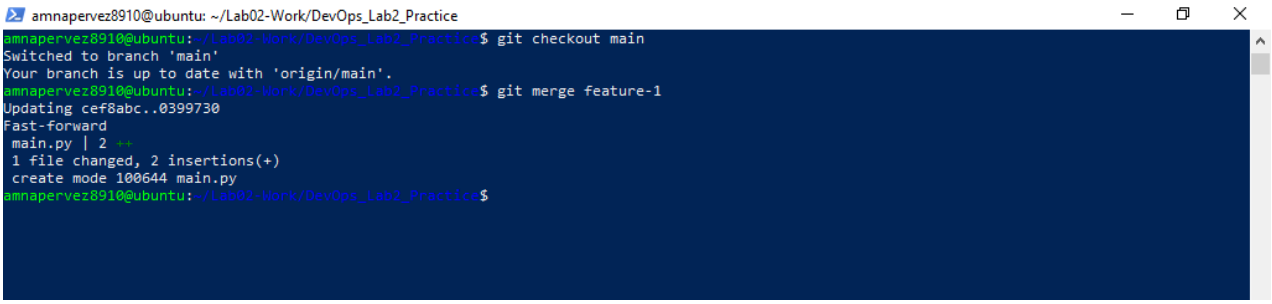
feature_commit.png



Greeting function added to main.py and committed on feature-1.

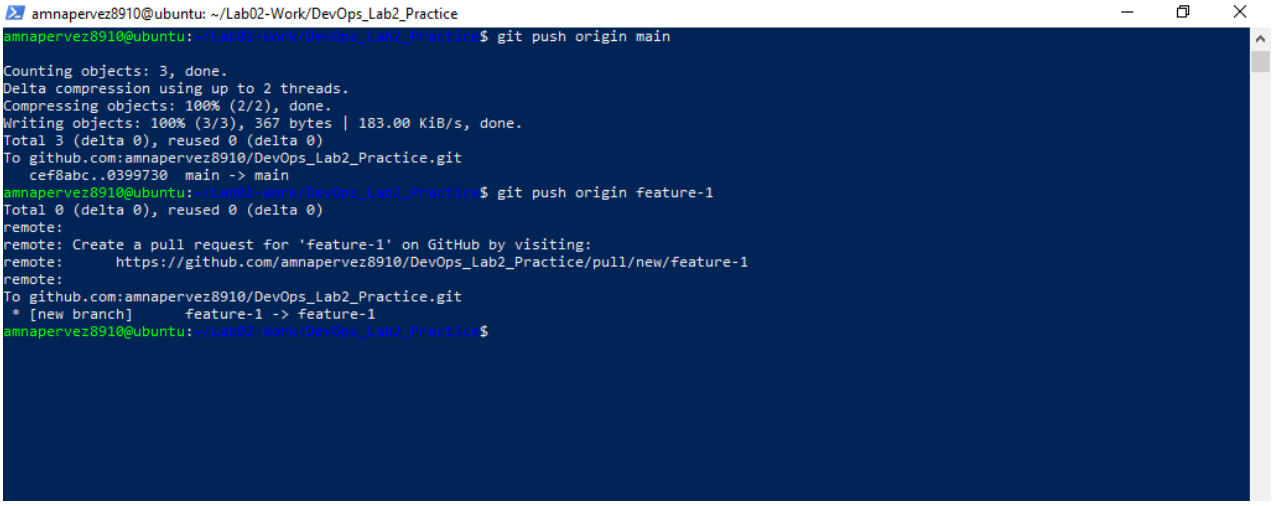
Task 9: Branching and Merging (continued)

merge.png



feature-1 merged into main using a fast-forward merge.

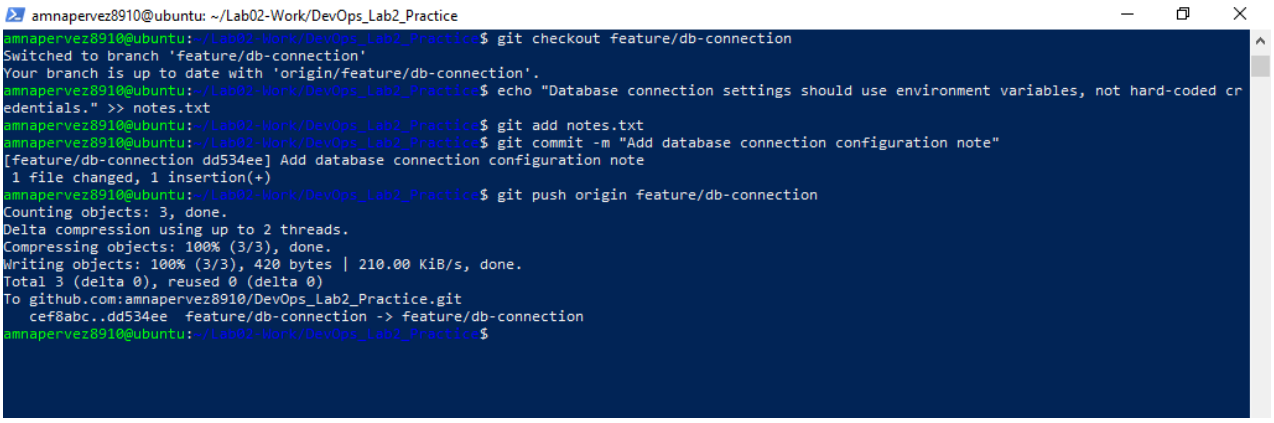
push_branches.png



main and feature-1 pushed to GitHub.

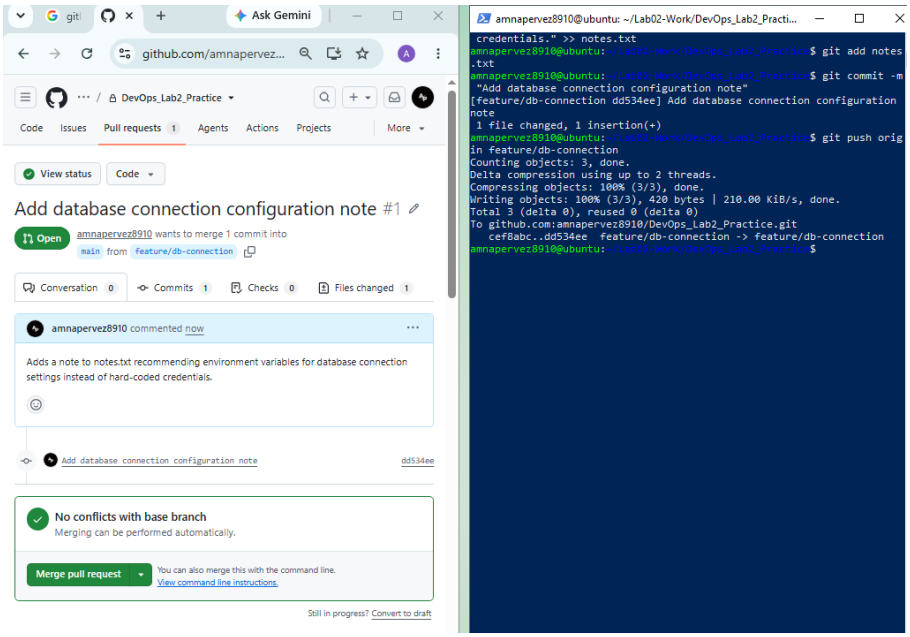
Task 10: Pull Request and Branch Review

db_connection_commit.png



Database configuration note committed and pushed on feature/db-connection.

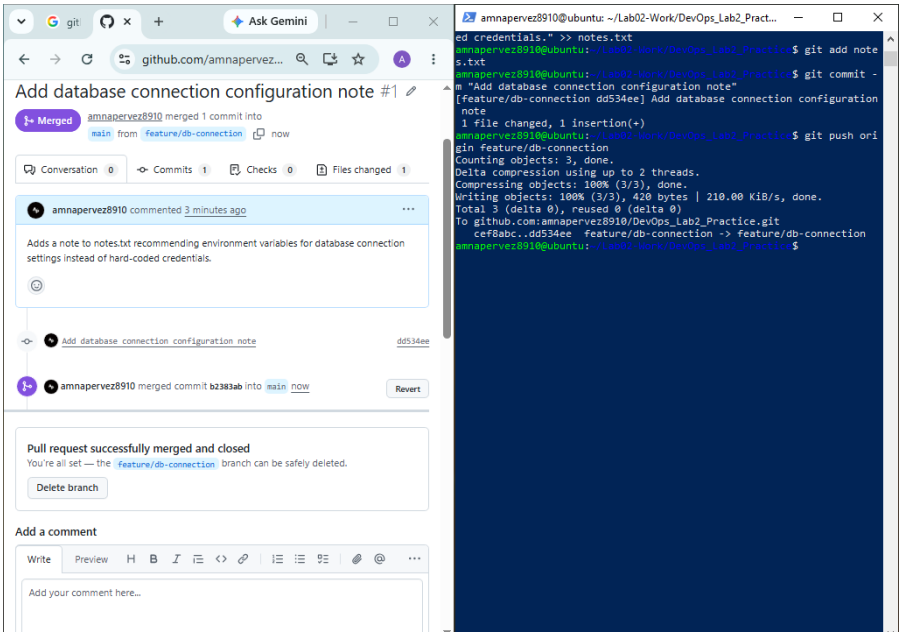
pr_creation.png



Pull request #1 created from feature/db-connection into main.

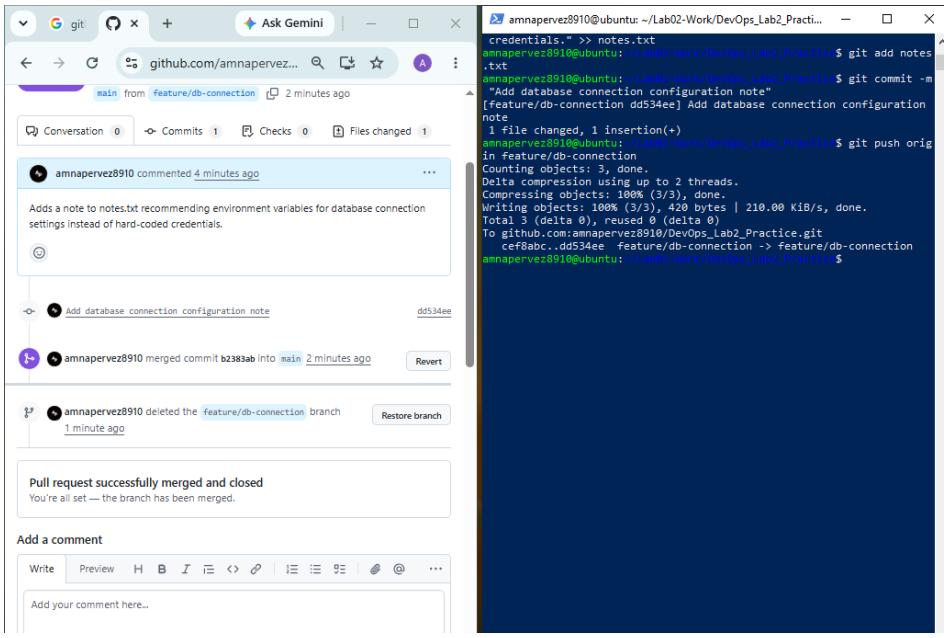
Task 10: Pull Request and Branch Review (continued)

pr_merge.png



Pull request #1 successfully merged and closed.

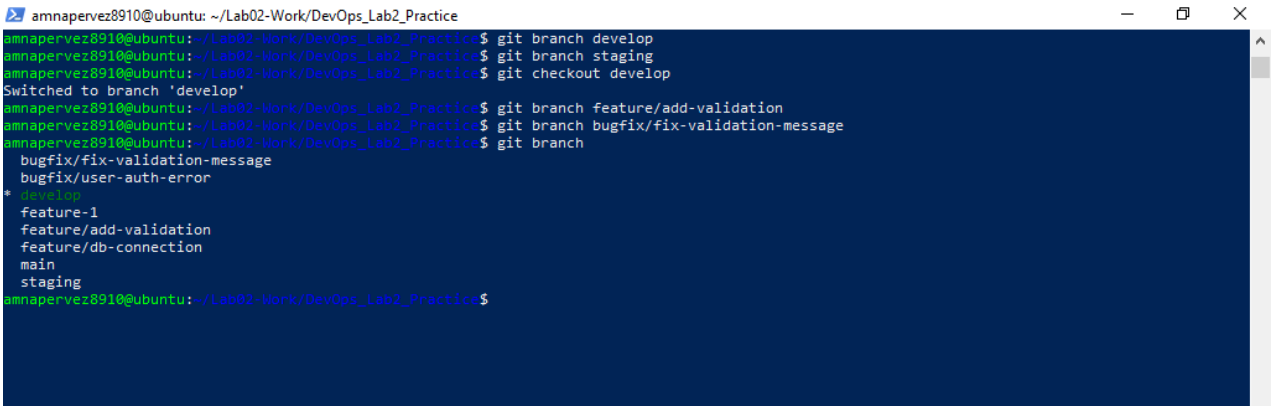
branch_delete.png



Merged feature/db-connection remote branch deleted through GitHub.

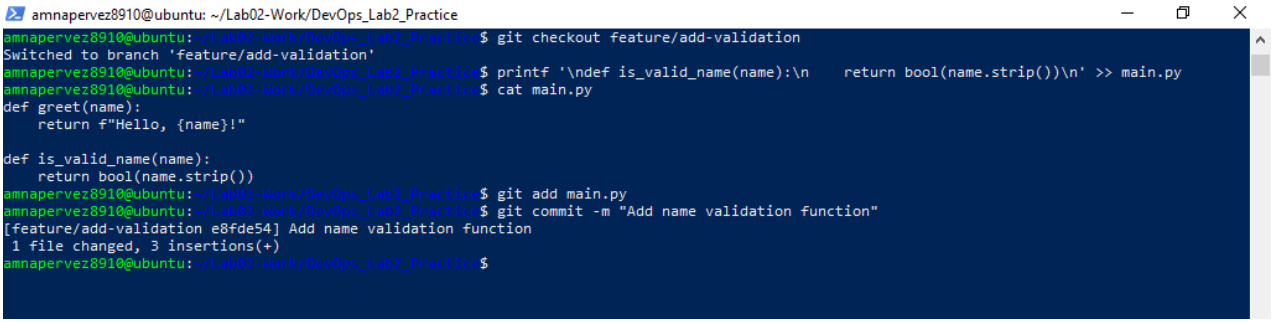
Task 11: Develop / Staging Strategy

branch_strategy.png



main, develop, staging, feature and bugfix branches listed in Ubuntu.

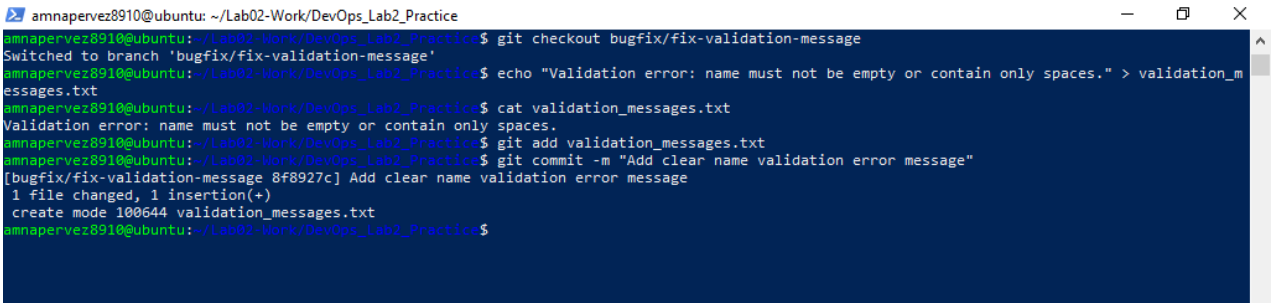
validation_feature_commit.png



Name-validation function committed on feature/add-validation.

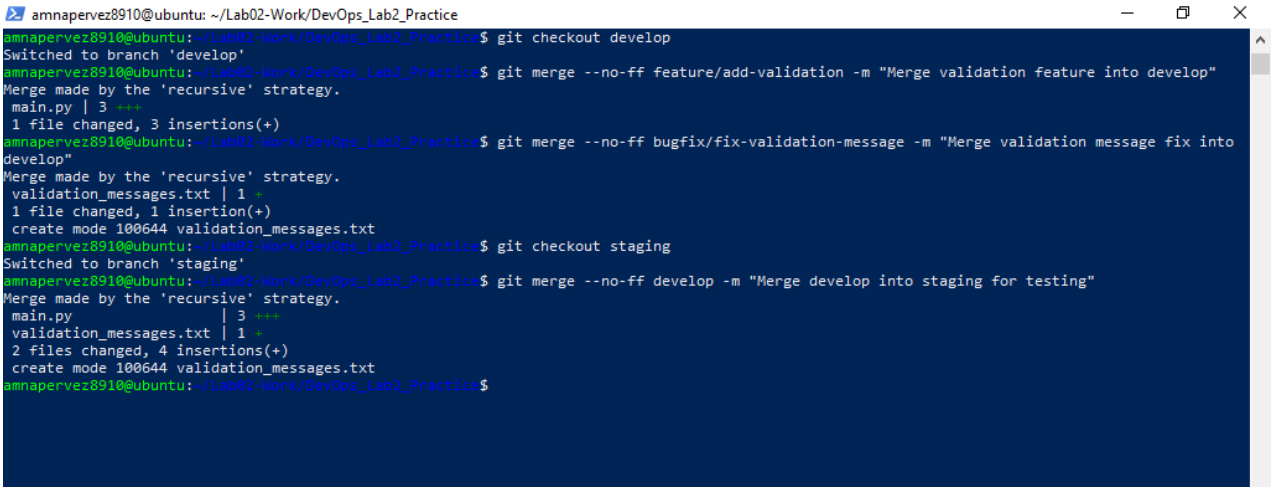
Task 11: Develop / Staging Strategy (continued)

validation_bugfix_commit.png



Clear validation message committed on bugfix/fix-validation-message.

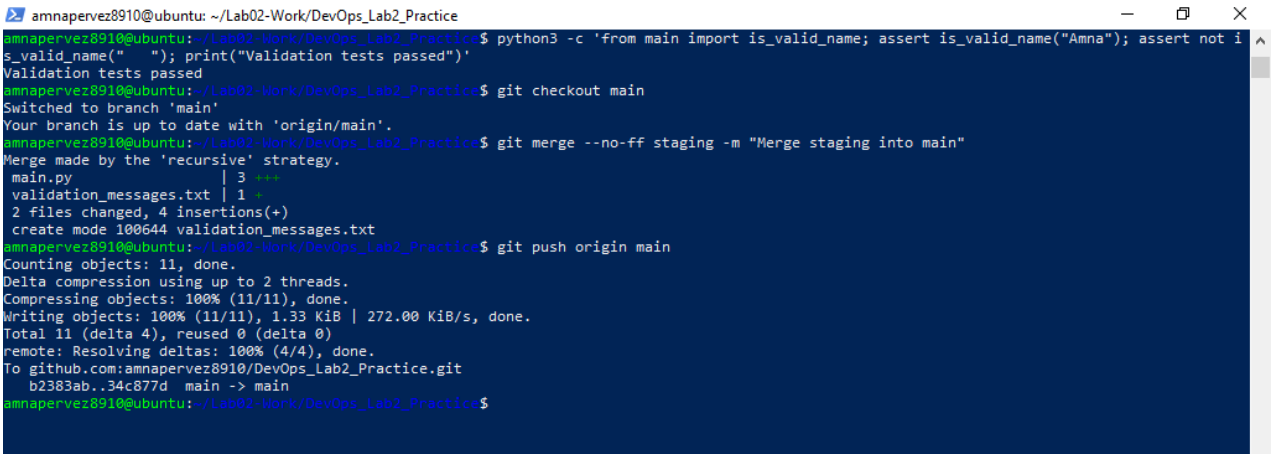
branch_merges.png



Feature and bugfix merged into develop, then develop merged into staging.

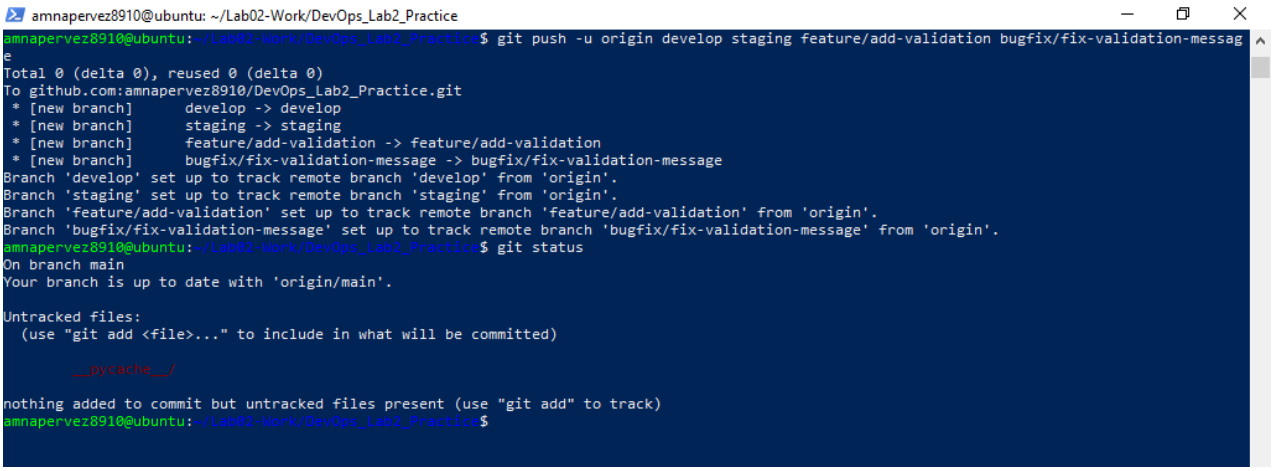
Task 11: Develop / Staging Strategy (continued)

final_merge.png



Validation checks passed; staging merged into main and pushed.

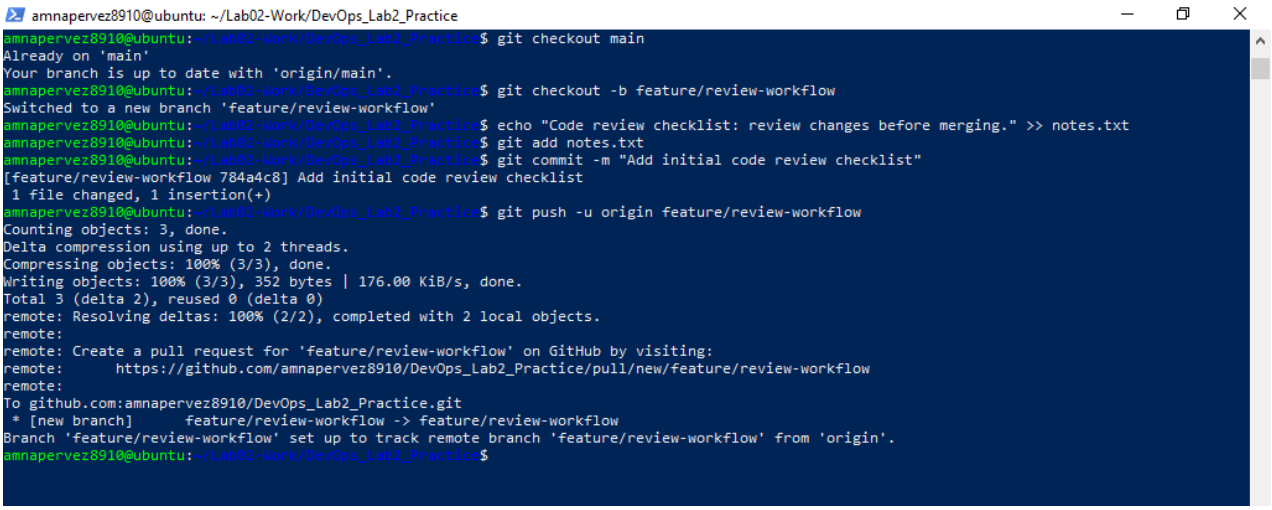
workflow_branches_push.png



Workflow branches pushed. Generated Python cache shown here was subsequently removed locally.

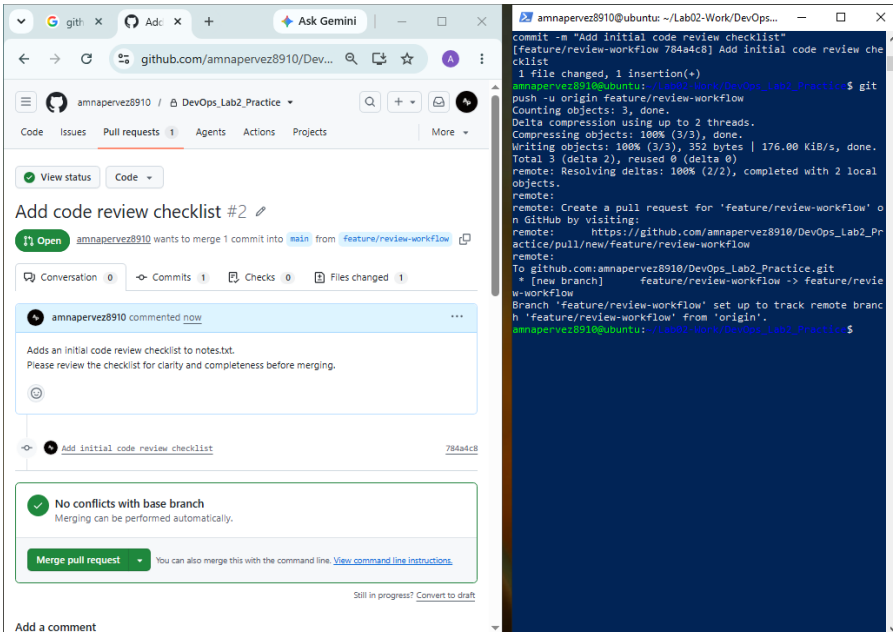
Task 12: Code Review Workflow

review_branch_push.png



Initial review checklist committed and pushed on feature/review-workflow.

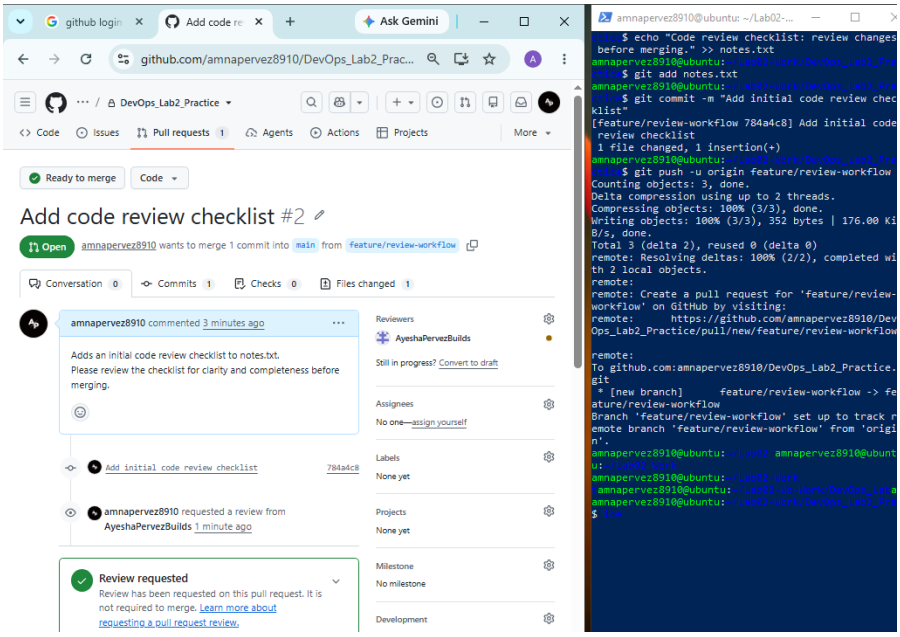
pr_create_details.png



Pull request #2 created with a clear title and description.

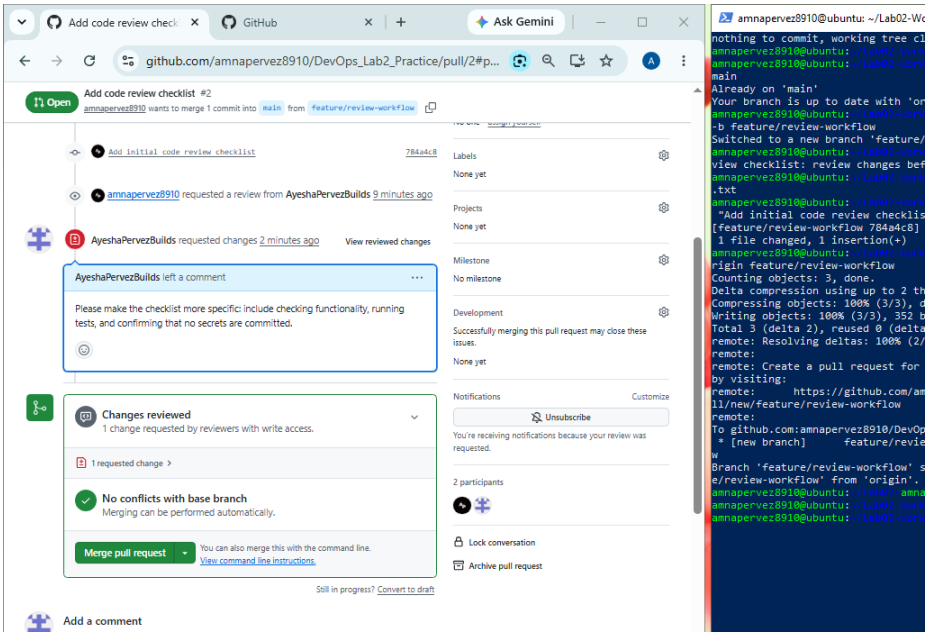
Task 12: Code Review Workflow (continued)

pr_assigned_reviewer.png



AyeshaPervezBuilds assigned as reviewer; review request visible.

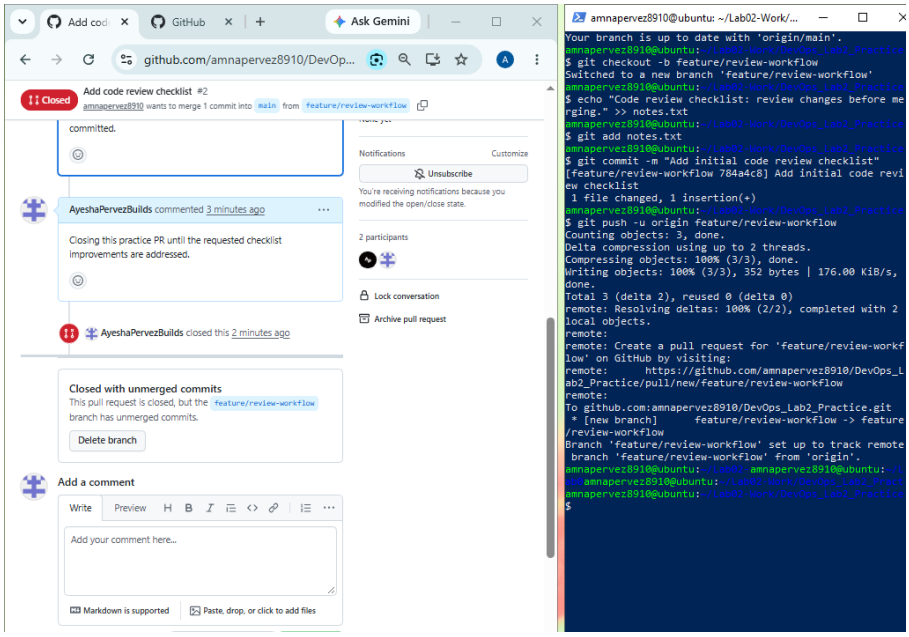
pr_request_changes.png



Reviewer requested specific functionality, testing and secret checks.

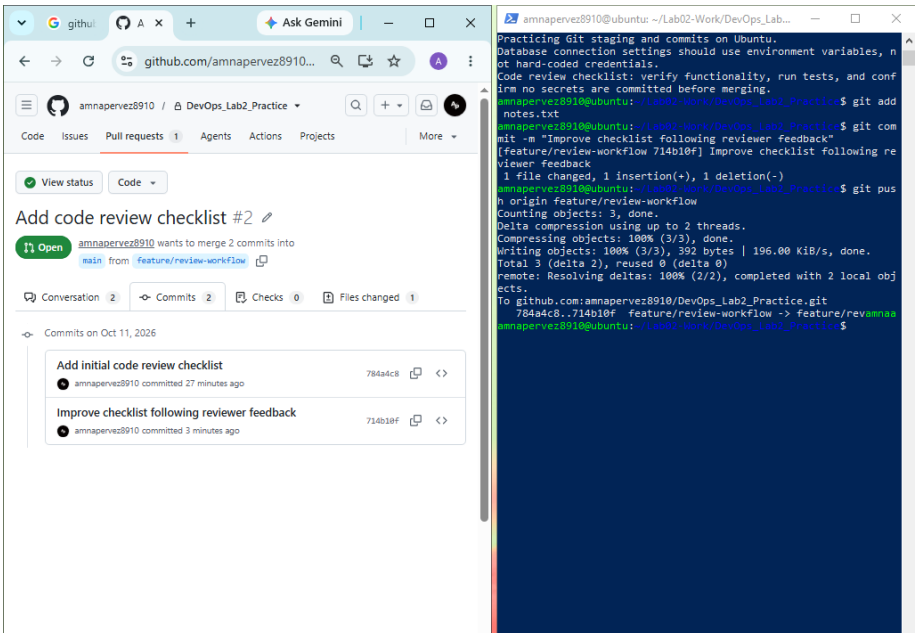
Task 12: Code Review Workflow (continued)

pr_rejected.png



Reviewer closed the practice PR pending improvements; the same PR was later reopened.

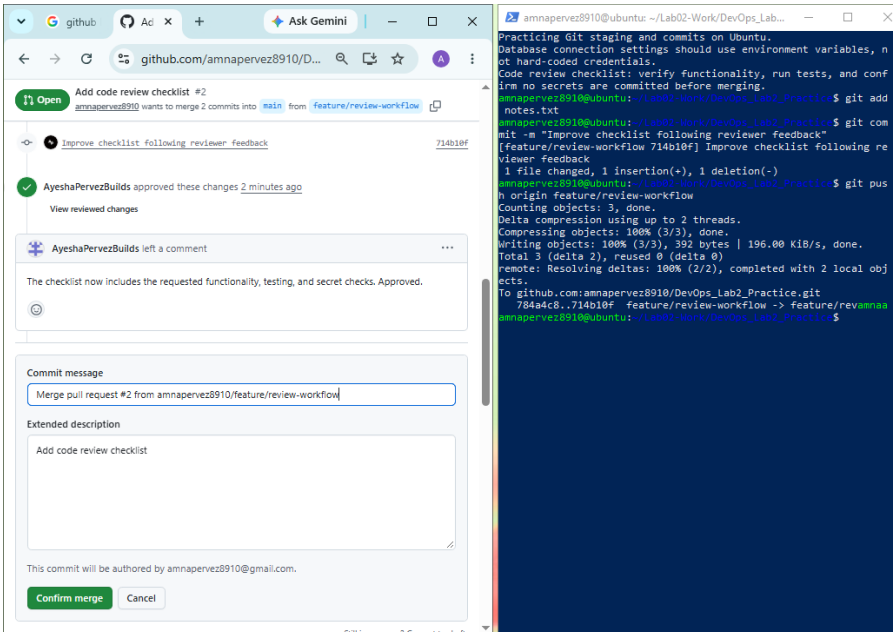
pr_updated_with_commits.png



Same PR updated with a second commit addressing reviewer feedback.

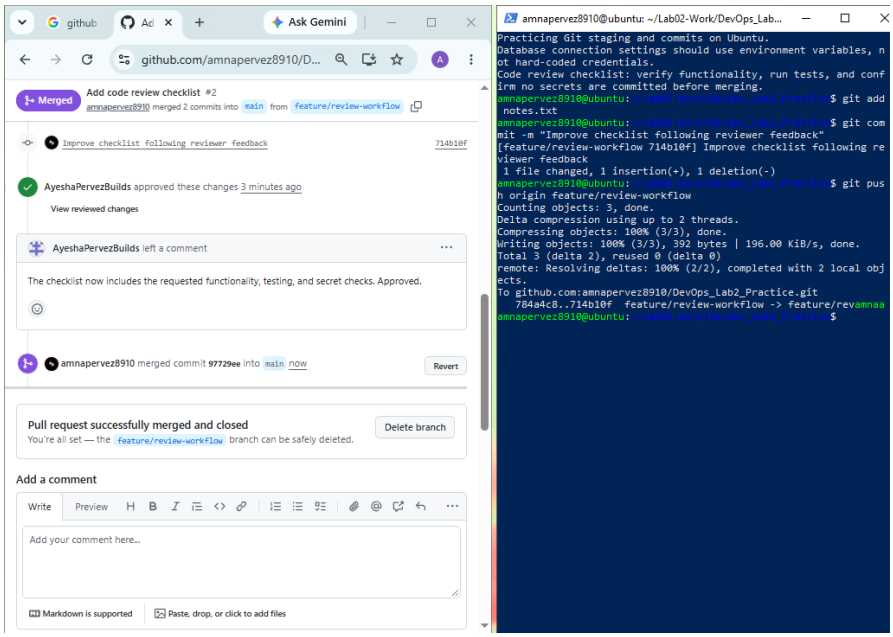
Task 12: Code Review Workflow (continued)

pr_merge_confirm.png



Reviewer approval and merge confirmation form shown before confirming the merge.

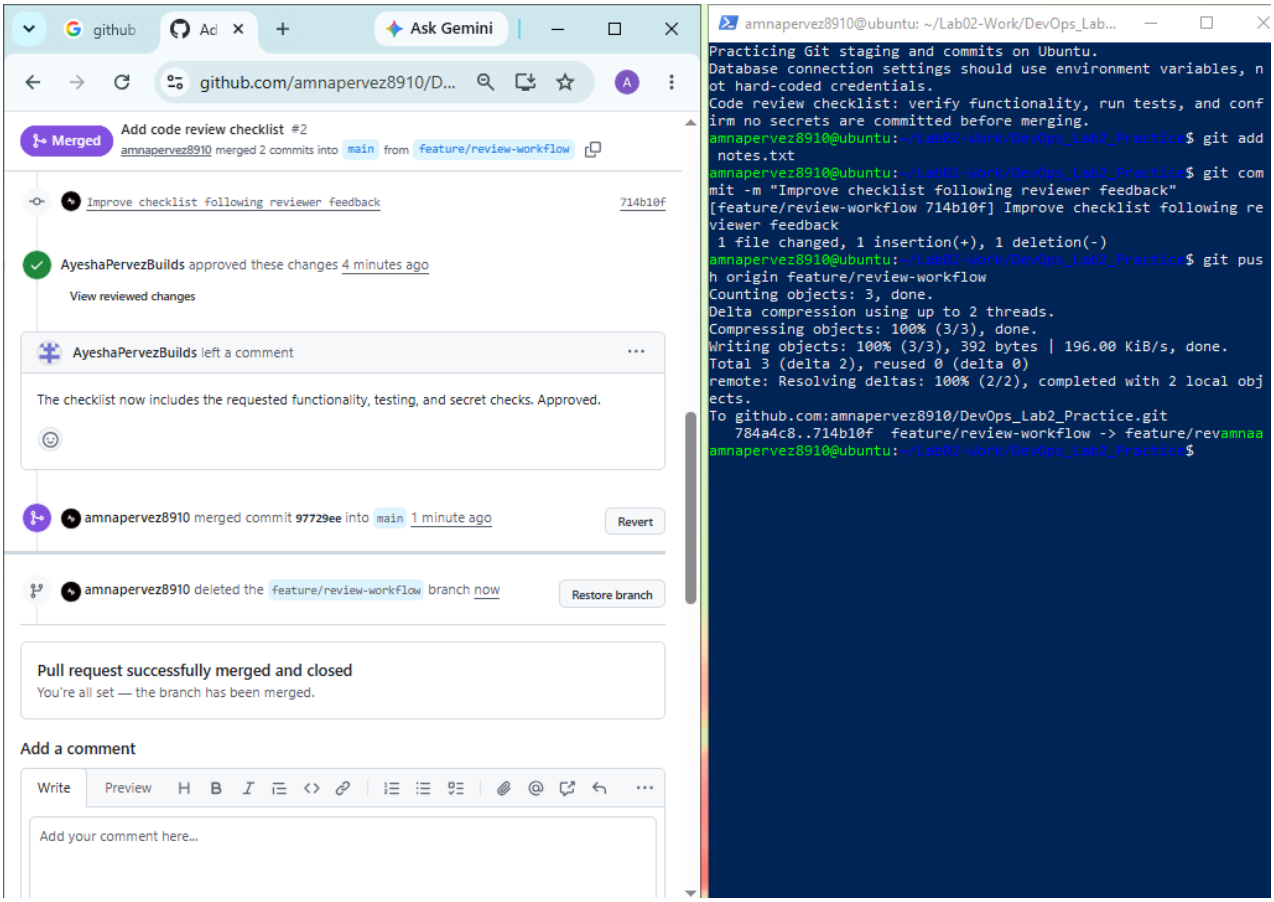
pr_merged.png



Approved pull request #2 merged and closed.

Task 12: Code Review Workflow (continued)

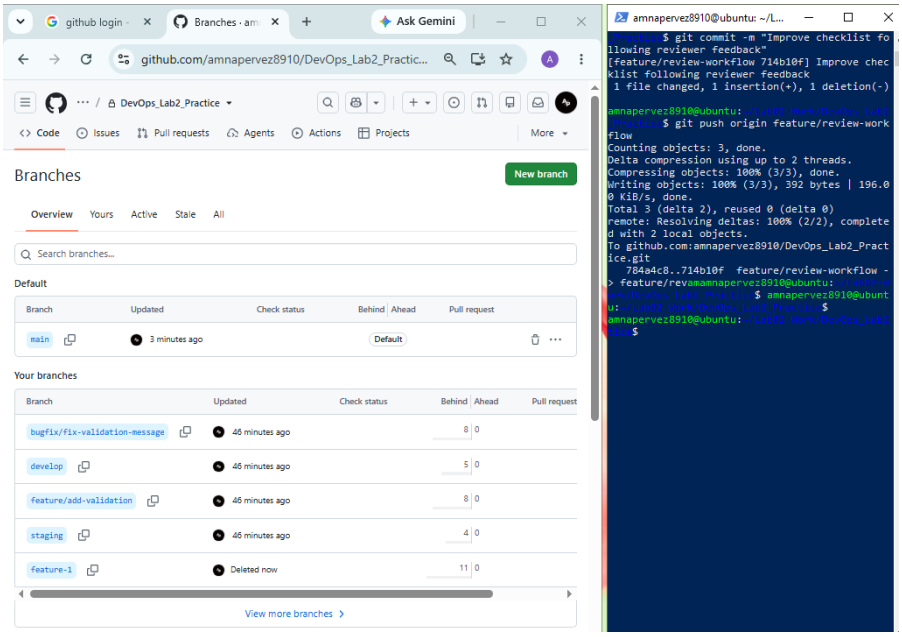
pr_branch_deleted.png



Merged feature/review-workflow remote branch deleted.

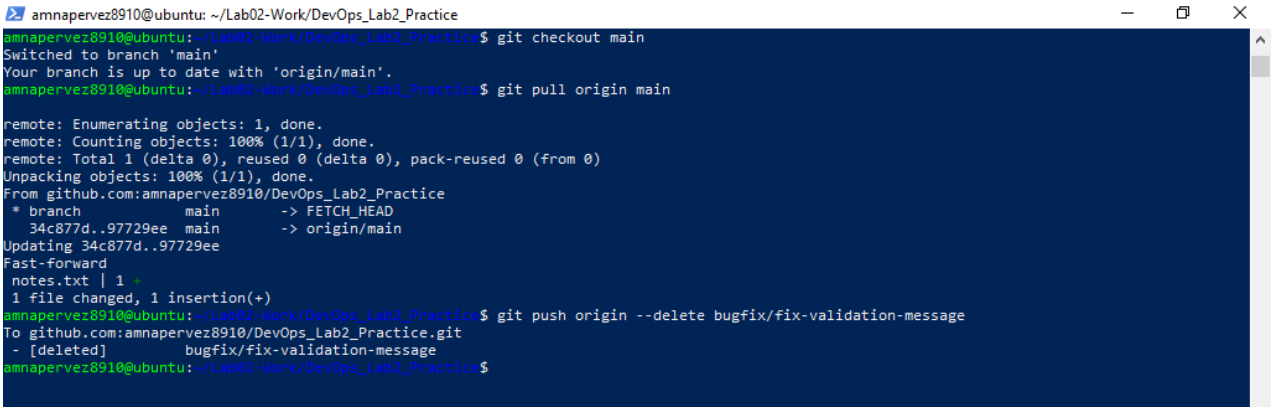
Task 13: Branch Cleanup

remote_branch_deleted.png



Merged feature-1 remote branch deleted through the GitHub interface.

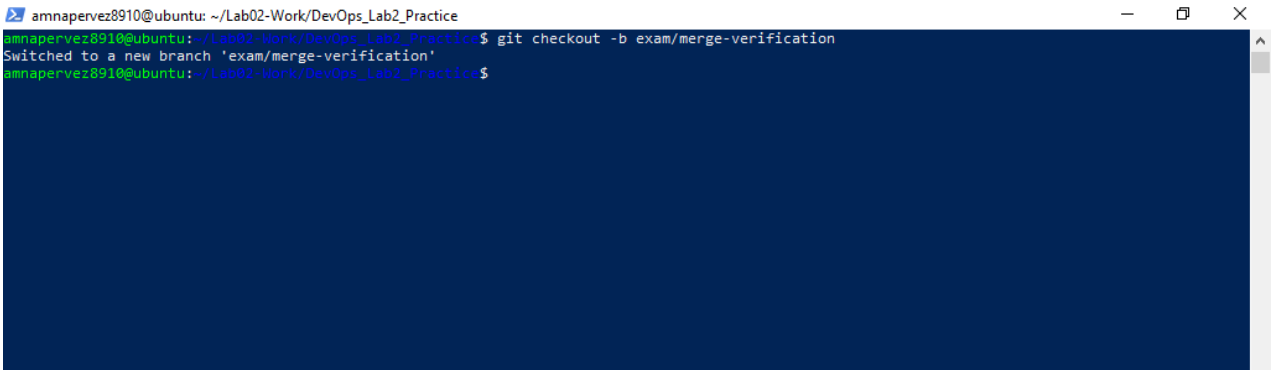
remote_branch_delete_cmd.png



main synchronized; completed bugfix remote branch deleted from Ubuntu.

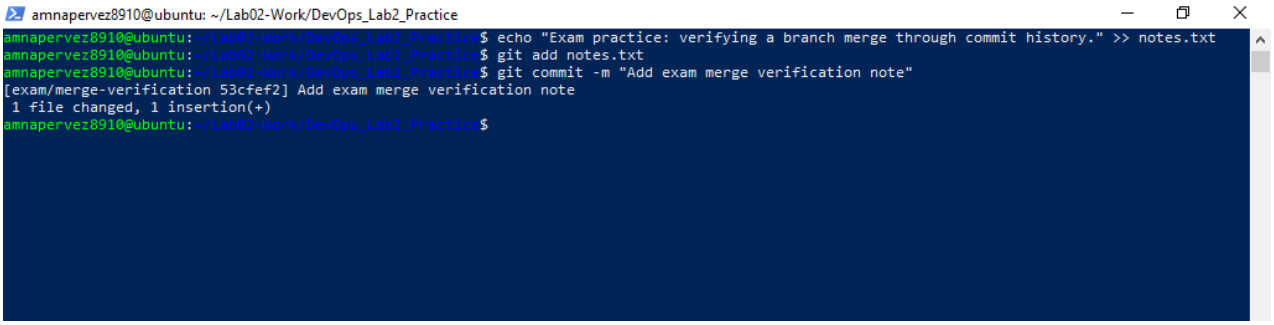
Q1: Advanced Branching and Verification

Q1_branch_created.png



exam/merge-verification branch created.

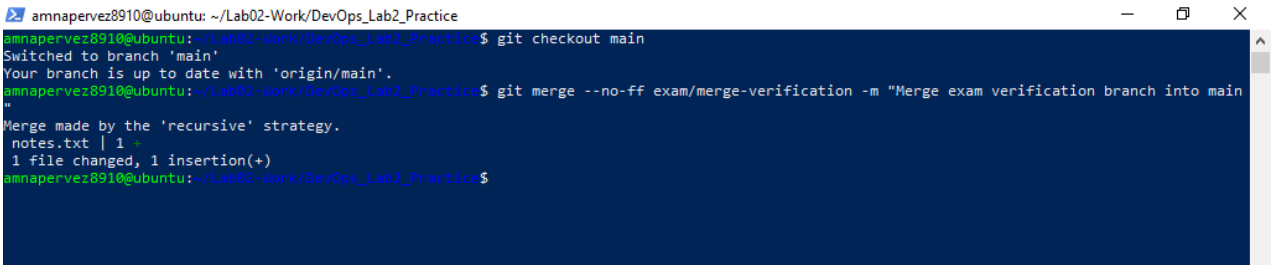
Q1_commit_done.png



Exam verification note added to notes.txt and committed.

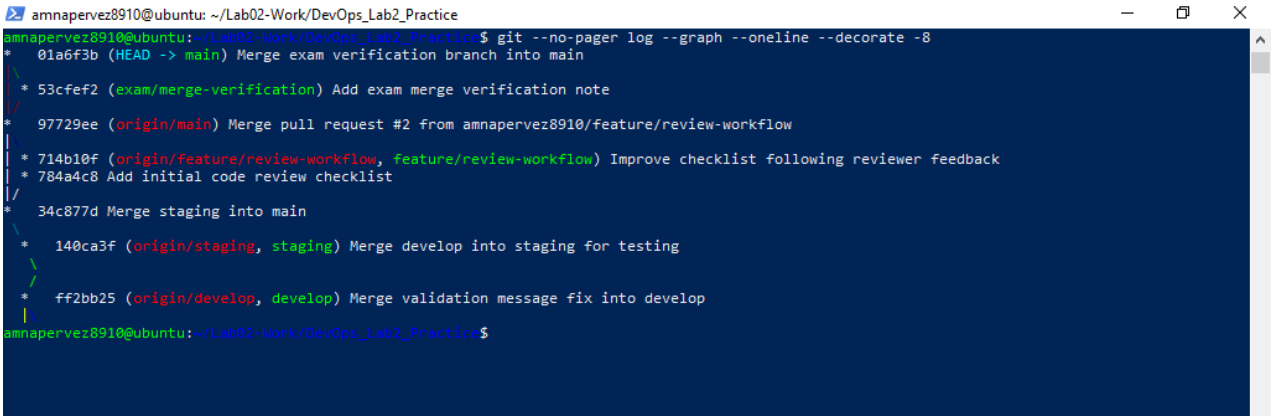
Q1: Advanced Branching and Verification (continued)

Q1_merge_done.png



Exam branch merged into main with an explicit merge commit using --no-ff.

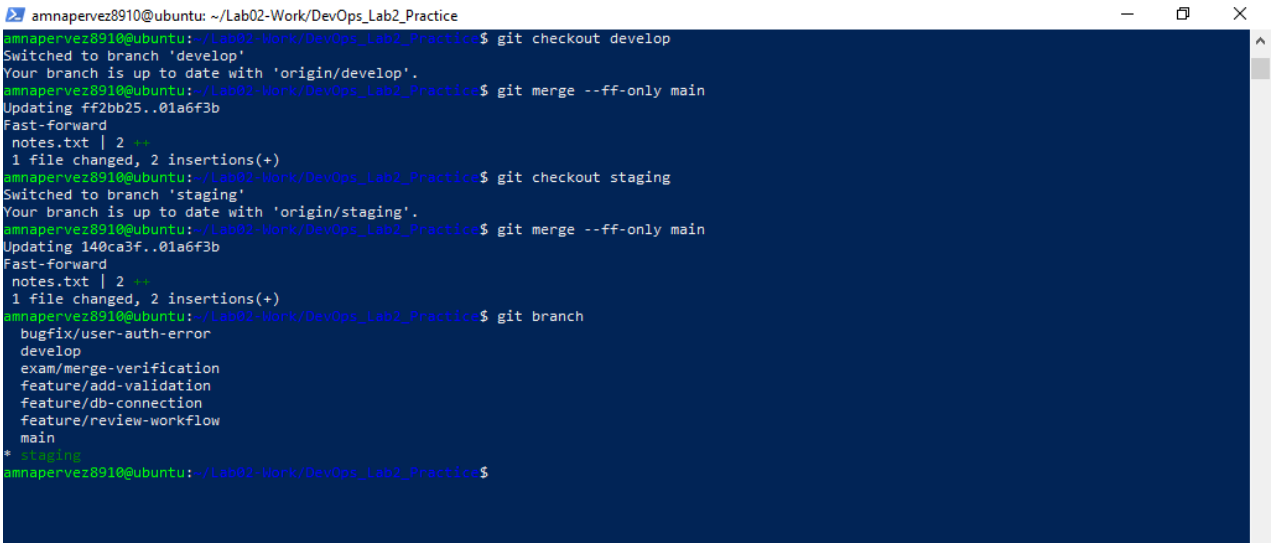
Q1_merge_history.png



Commit graph verifies the exam commit and its merge into main.

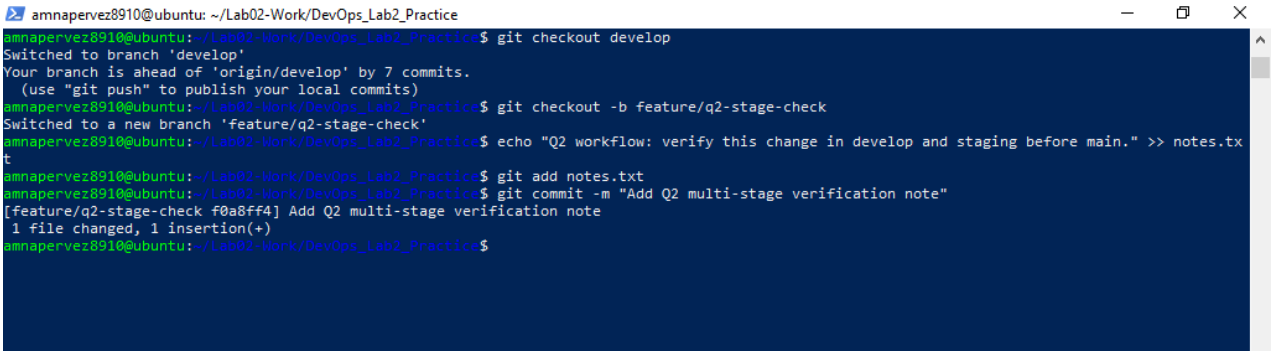
Q2: Multi-Stage Workflow

Q2_branches_created.png



Existing main, develop and staging branches reused; develop and staging synchronized to the starting baseline.

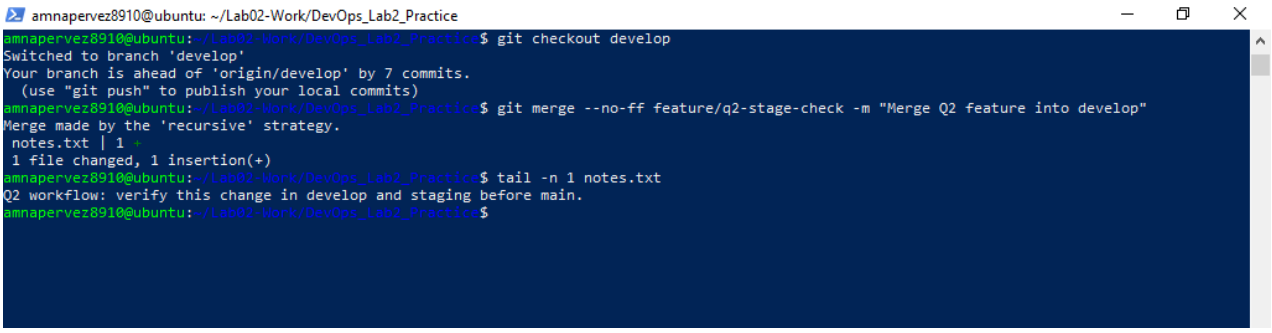
Q2_feature_branch_commit.png



Q2 feature branch created from develop; workflow note committed.

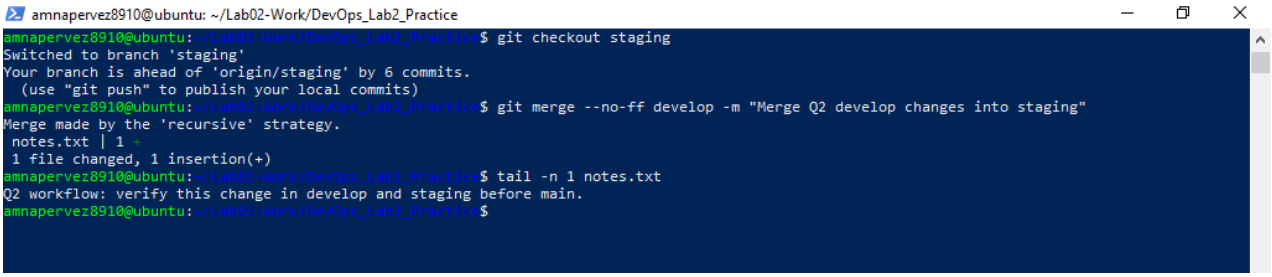
Q2: Multi-Stage Workflow (continued)

Q2_merge_into_develop.png



Feature merged into develop; updated note verified.

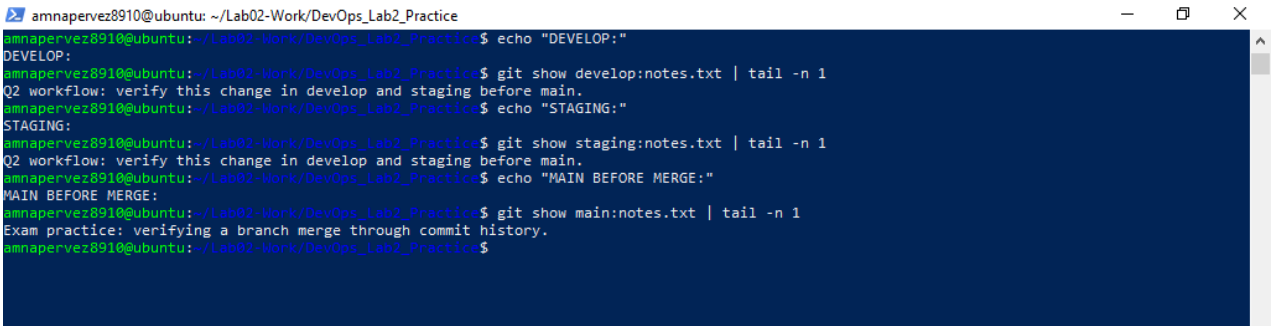
Q2_merge_into_staging.png



develop merged into staging; updated note verified.

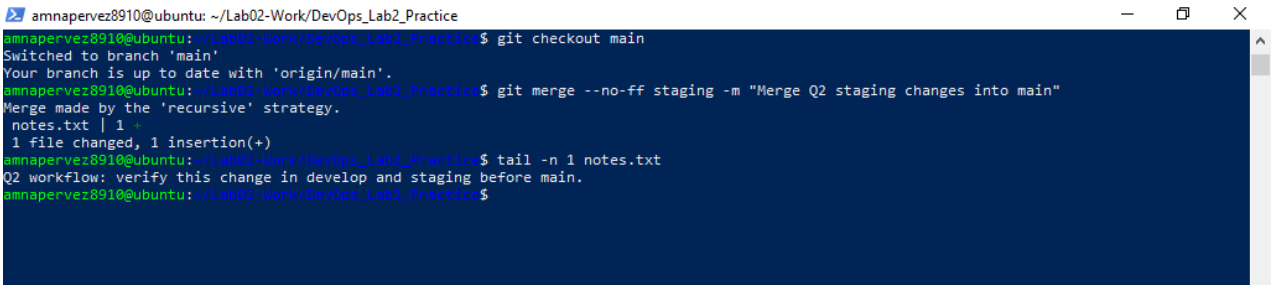
Q2: Multi-Stage Workflow (continued)

Q2_stage_verification.png



Updated note present in develop and staging but not yet in main, before promotion.

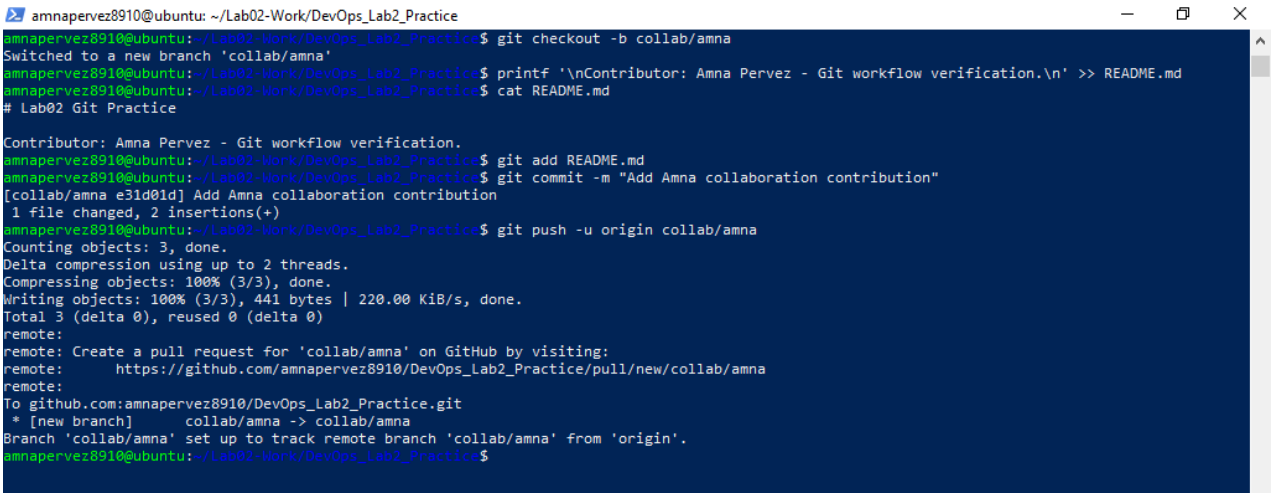
Q2_merge_into_main.png



staging merged into main; updated note verified in main.

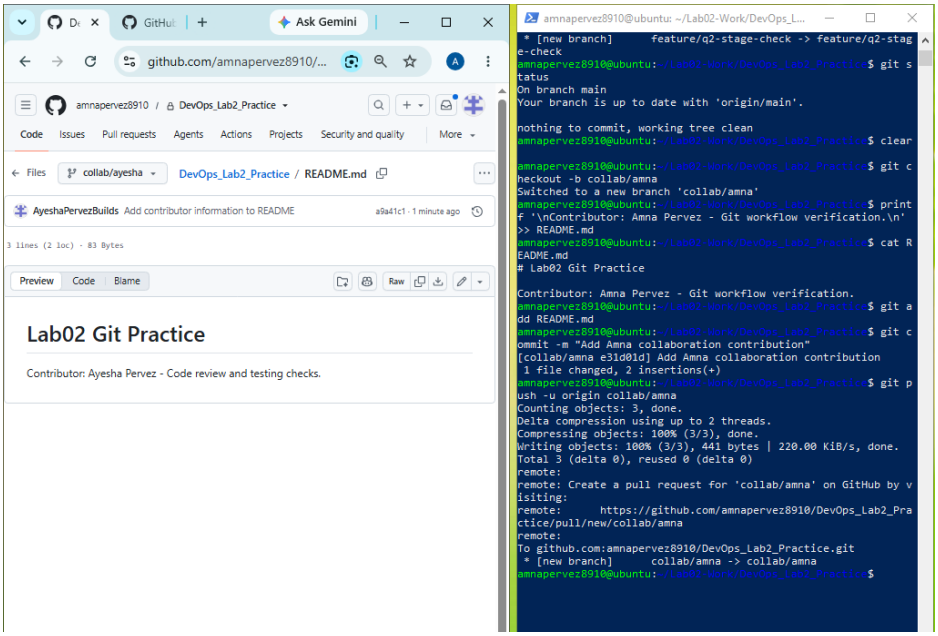
Q3: Collaboration and Conflict Resolution

Q3_branches_modified1.png



Amna contribution committed and pushed on collab/amna.

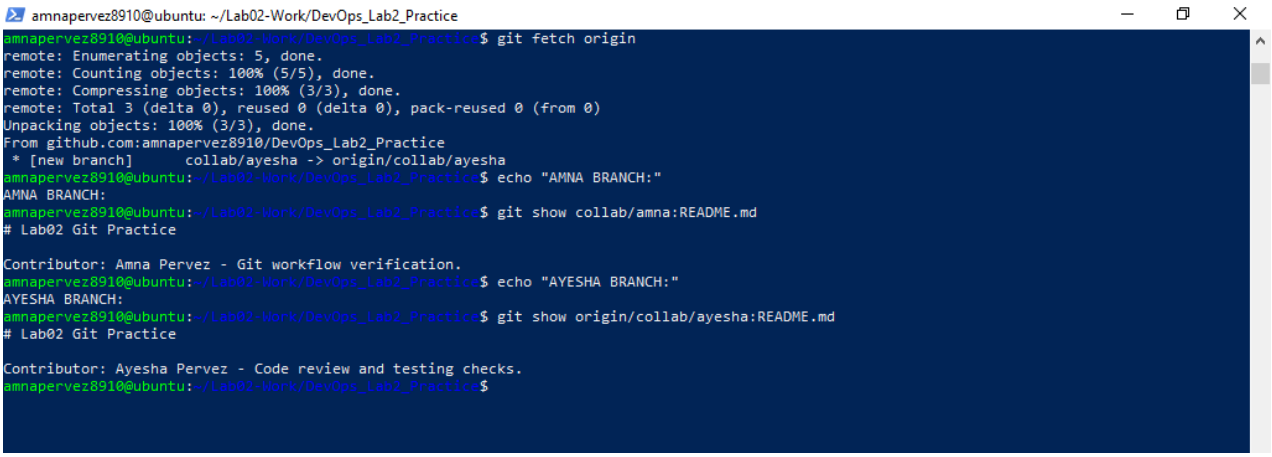
Q3_branches_modified2.png



Ayesha contribution committed through her GitHub account on collab/ayesha.

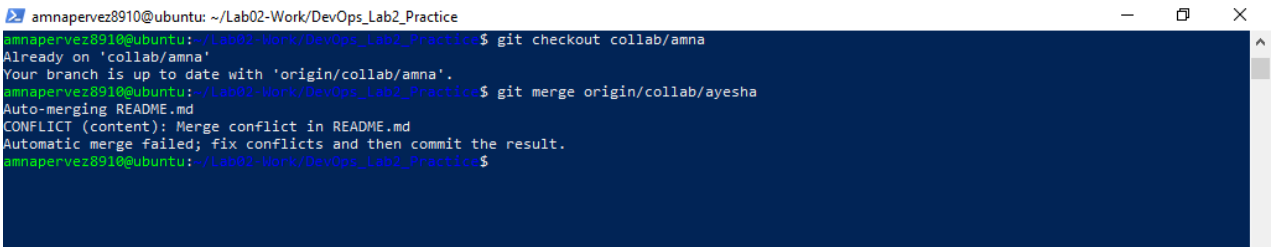
Q3: Collaboration and Conflict Resolution (continued)

Q3_branches_modified3.png



Both branch versions fetched and compared before merging.

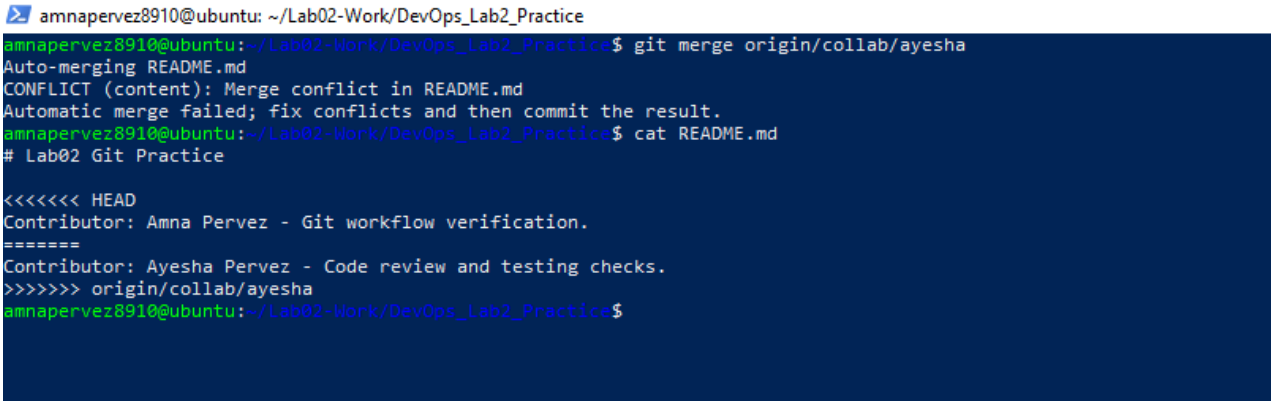
Q3_conflict_message.png



Attempted branch merge reports the planned README.md conflict.

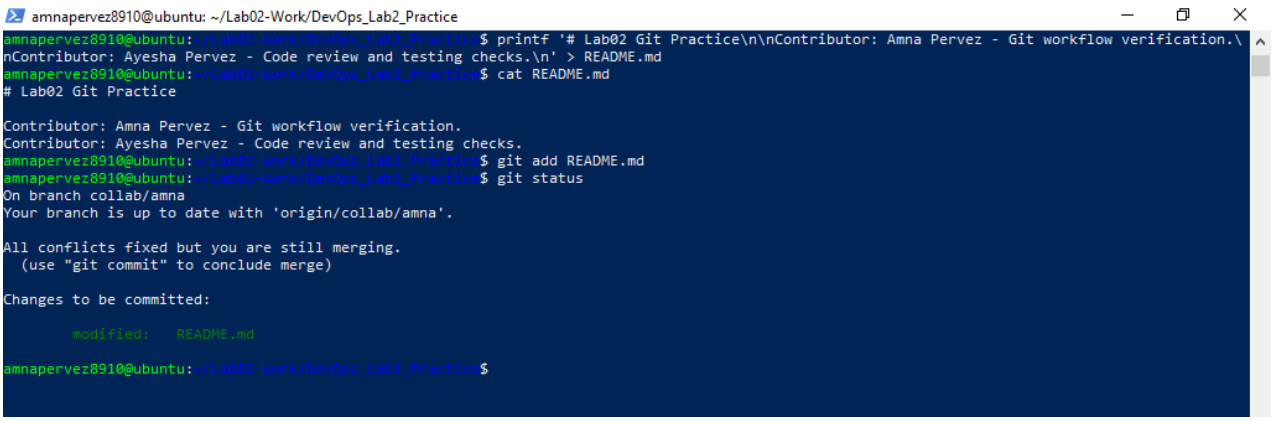
Q3: Collaboration and Conflict Resolution (continued)

Q3_merge_conflict.png



Conflict message and README conflict markers shown with both contributions.

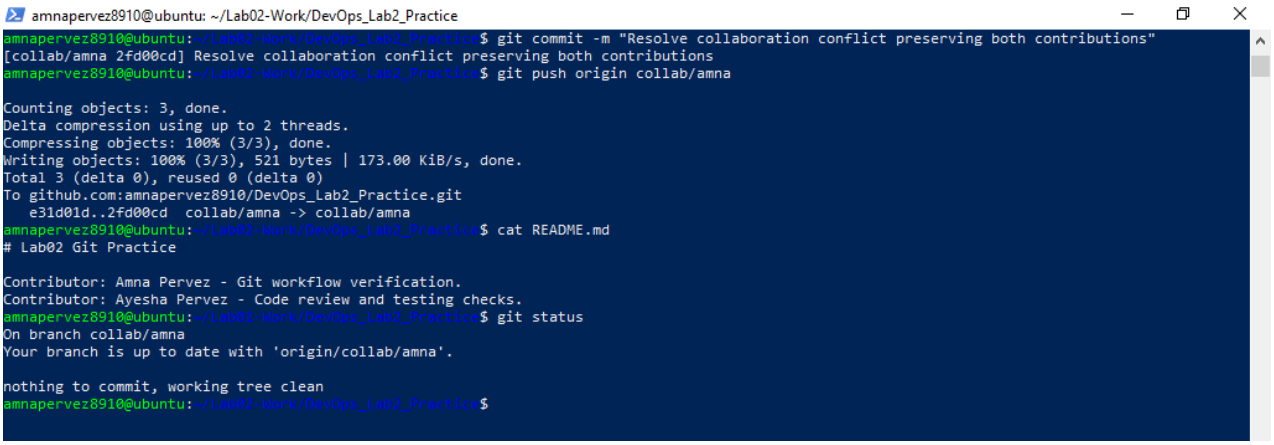
Q3_conflict_resolved.png



Conflict markers removed; both contributions preserved and file staged.

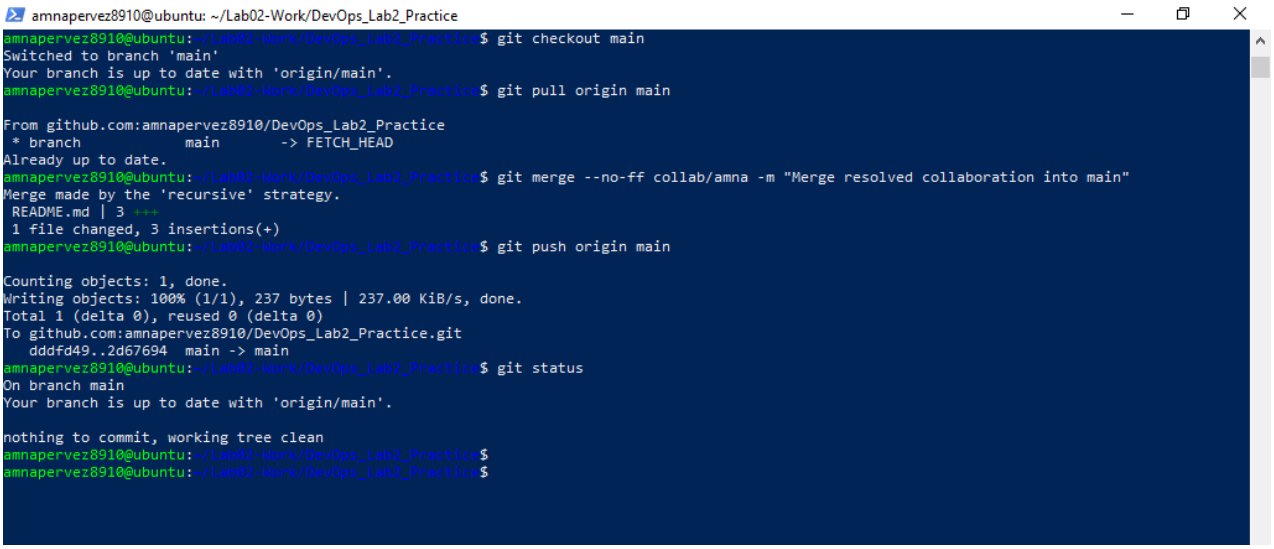
Q3: Collaboration and Conflict Resolution (continued)

Q3_final_file.png



Conflict-resolution commit pushed; final README contains both contributors and working tree is clean.

Q3_final_merge.png



Resolved collaboration merged into main, pushed, and clean status verified.